MIL-M-49233A(CR)
29 April 1988
SUPERSEDING
MIL-M-49233(CR)
16 November 1981

MILITARY SPECIFICATION

MODEM, CABLE DRIVER, MD-1025/G

This specification is approved for use within the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, fabrication and testing necessary for the manufacture of the Modem, Cable Driver MD-1025/G. This unit is also known as the Remote Loop Group Multiplexer Cable Driver (RLGM/CD) and is one of the electronic units that comprises the Digital Group Multiplexer (DGM) family of cable system equipment developed by the Joint Tactical Command, Control and Communications Agency (JTC$^3$A, formerly TRI-TAC). See 6.1 for intended use and functional description of the unit.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document shall be addressed to Commander, US Army Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey 07703-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC No. N/A FSC 5820

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

## 2. APPLICABLE DOCUMENTS

### 2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-P-11268 | Parts, Materials, and Processes Used in Electronic Equipment |
| MIL-M-13231 | Marking of Electronic Items |
| MIL-F-14072 | Finishes for Ground Electronic Equipment |
| MIL-C-55583 | Cable Assembly, Special Purpose Electrical CX-11230( )/G Cable Assembly Adapter CX-10734( )/G |
| MIL-E-55585 | Electronics Equipment and Parts, Packaging of |

STANDARDS

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-252 | Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices |
| MIL-STD-454 | Standard General Requirements for Electronic Equipment |
| MIL-STD-461A Notice 4 | Electromagnetic Interference Characteristics Requirements for Equipment, Subsystem and System |
| MIL-STD-462 Notice 3 | Electromagnetic Interference Characteristics, Measurement of |
| MIL-STD-781C | Reliability Tests, Exponential Distribution |
| MIL-STD-810D | Environmental Test Methods and Engineering Guidelines |

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| MIL-STD-1332 | Tactical Prime, Precise and Utility Terminologies for Classification of DOD Mobile Electronic Power Engine Generator Set Family |
| MIL-STD-2073-2 | Packaging Requirement Codes |
| MIL-STD-45662 | Calibration System Requirements |

2.1.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

SPECIFICATIONS

Communications-Electronics Command

| | |
|---|---|
| CR-CX-0173-001 | Nuclear Survivability.Vulnerability Provision for the Family of Digital Group Multiplexer Equipment |

DRAWINGS

Communications-Electronics Command

| | |
|---|---|
| DL-SM-B-875930 | Modem, Cable Driver, MD-1025()/G |
| SM-D-875948 | Circuit Card Assembly (CCA), BITE 1 |
| SM-D-875954 | CCA, Cable Driver |
| SM-D-876258 | CCA, Cable Interface |
| SM-D-884567 | CCA, Multi-Vdc |
| SM-D-884579 | CCA, AC Input |
| SM-D-884602 | CCA, 180/180 Vdc |

(Copies of specifications, standards, drawings, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

## 3. REQUIREMENTS

3.1 First article units. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.

3.2.1 Fabrication. The equipment shall be constructed in accordance with the requirements of this specification and of DL-SM-B-875930, including the requirements for parts, materials and processes therein.

3.2.2 Printed wiring. The printed wiring shall meet the requirements of MIL-STD-454, Requirement 17.

3.2.3 Semiconductor devices and integrated circuits. The semiconductor devices and integrated circuits shall meet the requirements of MIL-STD-454, Requirements 30 and 64.

3.3 Parts, materials, and processes - general. (See 4.4) In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and drawing DL-SM-B-875930.

3.5 Marking. The marking shall conform to MIL-M-13231, with approvals in accordance with requirements specified in the contract or purchase order.

3.5.1 Serial numbers. A unique serial number is required on each nomenclatured item and CCA. They shall be consecutively assigned.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

3.6 Electrical preconditioning (burn-in). (See 4.6) Prior to submission to the specified quality conformance inspections, each assembled unit shall be electrically preconditioned for a minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free. Each unit fabricated shall be subjected to this process. Failure free is defined as (1) no built-in test equipment (BITE) alarms during burn-in, and (2) the successful completion of the operational tests of 4.7.7 after burn-in.

3.7 Electrical. (See 4.7)

3.7.1 Balanced equipment-side signal characteristics. (See 4.7.1)

3.7.1.1 Line-to-line voltage levels. (See 4.7.1.1) The line-to-line voltage shall be between +0.1 and +4.0 volts for a digital 1, and between -0.1 and -4.0 volts for a digital 0, measured into a 100-ohm dummy load.

3.7.1.2 Line-to-ground voltage. (See 4.7.1.2) The magnitude of the line-to-ground voltage shall not exceed 3 volts.

3.7.1.3 Clock/data phasing. (See 4.7.1.3) For those data signals having a related clock signal, the time difference between the upward transition of the clock signal and either the upward or downward transition of the data signal shall be no more than $\pm$240 ns for group rate signals. Times shall be measured at the zero voltage point of the signals.

3.7.1.4 Offset. (See 4.7.1.4) The magnitude of the offset voltage measured between the center tap of a 100-ohm dummy load and ground shall not be greater than 3 volts.

3.7.1.5 Signal balance. (See 4.7.1.5) The magnitude of the difference between the magnitudes of the digital 1 level and the digital 0 level shall be less than 10% of the larger of the two.



Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

3.7.1.6 Slew rate. (See 4.7.1.6) When 100 feet of RG-108 A/U cable terminated in a 78-ohm dummy load is connected to the driver output, the transition between ±40 mV (either direction) shall take no more than 10 ns for group rate signals.

3.7.1.7 Receiver dynamic range. (See 4.7.1.7) The receiver shall deliver the proper digital output when a line-to-line voltage of +50 mV to +4V for a digital 1, or -50 mV to -4V for a digital 0, is applied to the receiver input terminals.

3.7.1.8 Input load impedance. (See 4.7.1.8) The input impedance shall be 100 ohms ±10%.

3.7.1.9 Clock asymmetry. (See 4.7.1.9) The positive half of the clock waveform shall be equal to half the period ±120 ns for group rate timing signals.

3.7.2 Cable-side group signal characteristics. (See 4.7.2)

3.7.2.1 Output voltage levels. (See 4.7.2.1) The diphase group output voltage shall be 3 volts ± 20% peak-to-peak into a 58-ohm resistive load.

3.7.2.2 Output rise and fall time. (See 4.7.2.2) The diphase group output rise and fall time shall be 130 ns maximum into a 58-ohm resistive load.

3.7.2.3 Output impedance. (See 4.7.2.3) The diphase group output impedance shall be 58 ohms ± 15% at 72 and 144 kilobits per second (kb/s) and 50 to 100 ohms from 300 to 1000 Hz.

3.7.2.4 Input impedance. (See 4.7.2.4) The diphase group input impedance shall be 58 ohms ± 15% from 1000 Hz to 300 kHz and 50 to 100 ohms from 300 to 1000 Hz.

3.7.2.5 Receiver sensitivity. (See 4.7.2.5) The unit shall operate properly on the signal specified in 3.7.1 at the end of 3.2 km of CX-11230 (see MIL-C-55583) cable with an additional 4 dB of attenuation. Proper operation shall be defined as a channel bit error rate (BER) of zero errors in 30 seconds.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

3.7.3 Equipment-side maintenance orderwire (MOW) signal characteristics. (See 4.7.3)

3.7.3.1 Input and output impedance. (See 4.7.3.1) The input and output impedance shall be from 540 to 660 ohms.

3.7.3.2 Output level. (See 4.7.3.2) Over a range of -5.5 to +4 dBm at 1 kHz applied to the cable side input, the output shall be -7 to -2 dBm into 600 ohms.

3.7.3.3 Frequency response. (See 4.7.3.3) The loss vs frequency characteristic of the orderwire channel shall be within the following limits:

| Frequency, Hz | Loss, dB |
|---|---|
| 300 | -5 to +8 |
| 400 | ±5 |
| 1000 | Reference |
| 3000 | ±5 |
| 3400 | -5 to +8 |

3.7.3.4 Signal-to-noise ratio. (See 4.7.3.4) With random group traffic the signal-to-noise ratio C-weighted shall be greater than 11 dB at 72 kb/s and 20 dB at 144 kb/s.

3.7.4 Cable-side maintenance orderwire (MOW) signal characteristics. (See 4.7.4)

3.7.4.1 Output power level. (See 4.7.4.1) With -4 dBm at 1 kHz applied to the equipment-side input, the output level shall be +5 dBm ±1 dB into a 58-ohm dummy load.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

3.7.5 Power interface. (See 4.7.5)

3.7.5.1 Prime power interface. (See 4.7.5.1) The ac primary power source shall be 115V ±10% at frequencies of 50 Hz, 60 Hz, and 400 Hz ±5% single phase. The power input to the equipmemt shall not exceed 200 watts. The ac primary power shall conform to MIL-STD-1332 for Type I, Class 1 and 2, Mode I, II, and III generator sets.

3.7.5.2 Power feed output over-voltage. (See 4.7.5.2) The maximum open circuit voltage shall be 210 Vdc.

3.7.5.3 Power feed current limits. (See 4.7.5.3) The overcurrent protection circuitry shall trip at 420 milliamperes maximum. The output shall shut down and discharge the output voltage through a resistance of 5000 ohms or less when the power feed current drops below 10 mA. After undercurrent shutdown, manual restart shall be required.

3.7.5.4 Current protection. (See 4.7.5.4) The cable interface circuits shall withstand without damage to the equipment an injected current of 90 milliamperes on either center conductor of the CX-11230 cable.

3.7.6 BITE and alarms. (See 4.7.6) The Built-In Test Equipment (BITE) shall monitor signals in and out of the unit and internal circuits and power supply voltages. The BITE shall operate without interrupting traffic or interfering with normal system operation. Alarm conditions are removed automatically when proper operation is restored. Failures detected by the BITE circuitry shall cause relay contact closures accessible externally at a connector on the rear panel. The following outputs shall be available. Means shall be provided in the unit to check the operation of the alarms without interrupting the equipment operation.

a. S Alarm. The S alarm shall indicate any failure detected including loss of input power. This alarm also provides a visible indication on the front panel except when there is a loss of input power.

b. SA Alarm. The SA alarm shall indicate that the malfunction is within the unit. Loss of input cable signal shall inhibit the SA alarm. A loss of signal on one cable input shall suppress only the SA alarm originating in the equipment



Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

associated with that cable input. SA alarms associated with a nonfailed cable input shall not be suppressed. If the power switch is set to OFF, there shall be no SA alarm.

c. LS Alarm. The LS alarm shall indicate loss of signal coming in on either CX-11230 cable. If the unit loses power, the LS alarm assumes its normal state. This alarm inhibits the SA alarm on the failed group.

3.7.7 Operational tests . (See 4.7.7)

3.7.7.1 Group error check. (See 4.7.7.1) There shall be no errors for a measurement period of 10 seconds under the following conditions:

Power feed ON

Test tone on the orderwire

144 kb/s at 3.2 km (2 miles) of cable

3.7.7.2 Maintenance orderwire check. (See 4.7.7.2) The maintenance orderwire output level with -4 dBm at 1 kHz input shall be -7 to -2 dBm.

3.7.7.3 Output dc voltage check. (See 4.7.7.3) The output dc voltage shall be 180 Vdc ±5% into a 600-ohm load. The polarity shall be positive on both center conductors and negative on the shields.

3.7.7.4 Alarm monitor check. (See 4.7.7.4) No alarms shall be generated; that is, there will be open circuits on the S, SA, and LS alarm outputs. (A short indicates an alarm).

3.7.8 Full duplex signal detection and dynamic range. (See 4.7.8) The modem shall be capable of recovering data in a full duplex circuit from a conditioned diphase signal transmitted over 0 to 3.2 km of CX-11230 cable at both the 72 and 144 kb/s output rates, without power feed, with no errors, and with the maintenance orderwire signal present at -4 dBm. The diphase signal shall be transmitted error free over a minimum length cable of 6 meters and a maximum length cable of 3.2 km with 4 dB attenuation. The BER test shall be made with a pseudorandom sequence for a minimum test time of 10 seconds.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

3.7.9 Environmental operational tests. (See 4.7.9) Environmental operational tests shall be in accordance with 3.7.7.

3.8 Service conditions. (See 4.8) The operational test of 3.7.9 shall be performed before, during, and after each service condition test unless otherwise specified in the appropriate test paragraph.

3.8.1 Temperature. (See 4.8.1)

3.8.1.1 High temperature. (See 4.8.1.1) The equipment shall be operable without degradation in specified performance at ambient air temperatures as high as +63°C (+145°F), and shall withstand (non-operating) exposure to ambient air temperatures as high as +71°C (+160°F).

3.8.1.2 Low temperature. (See 4.8.1.2) The equipment shall be operable without degradation in specified performance at ambient air temperatures as low as -31.7°C (-25°F), and shall withstand exposure to ambient air temperatures as low as -57°(-70°F).

3.8.2 Humidity. (See 4.8.2) The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.8.3 Low pressure (altitude). (See 4.8.3) The equipment shall be operable without degradation in specified performance at altitudes up to 10,000 feet above sea level, and shall withstand air transportation at altitudes up to 40,000 feet above sea level.

3.8.4 Dust. (See 4.8.4) The equipment shall withstand, in both operating and non-operating conditions, the effect of dust particles that may accumulate within the shelter assembly or vehicle.

3.8.5 Salt fog. (See 4.8.5) The equipment shall be resistant to the corrosive effects of a salt-fog atmosphere.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

3.8.6 Fungus. (See 4.8.6) The equipment, in the assembled-and-ready-for-delivery condition, shall provide no nutrients in material, coating or contaminant form, or support fungal growth when tested in accordance with 4.8.6. Only inherently fungus-resistant grades of materials per requirement 4, MIL-STD-454, shall be used.

3.8.7 Vibration and shock. (See 4.8.7) The equipment shall withstand vibration and shock induced during vehicular transport as part of a mobile assemblage, over all types of roads and cross-country terrain, and shocks encountered during servicing.

3.9 Electromagnetic interference(EMI). (See 4.9) The unit shall meet the following emission and susceptibility requirements of MIL-STD-461A, Notice 4:

| | | | |
|---|---|---|---|
| CE02 | CS02 | RE02 | RS02 |
| CE03 | CS06 | RE02.1 | RS03 |
| CE04 | | | |
| CE05 | | | |

NOTES:

a. RE02.1 is applicable over the frequency range of 14 kHz to 1000 MHz.

b. RS03 is applicable over the frequency range of 10 kHz to 1000 MHz.

c. Conducted emission and susceptibility requirements are applicable to 60-Hz and 400-Hz power lines.

3.9.1 Bonds and grounds. (See 4.9.1) Bonding and grounding impedance shall not exceed 2.5 milliohms.

3.10 Reliability. (See 4.10) The unit shall have a specified Mean-Time-Between-Failure (MTBF), $O_o$, of 4000 hours when in the normal mode of operation.

3.11 Interchangeability. (See 4.11) Like units, assemblies, subassemblies, and replaceable parts shall meet Requirement 7 of MIL-STD-454.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

3.12 Unit subassembly electrical performance. (See 4.12) The unit subassemblies listed below shall meet the requirements specified on their respective test requirement specification (TRS) drawings, as referenced in the following drawings:

| CCA | DRAWING |
|---|---|
| BITE 1 * | SM-D-875948 |
| Cable Driver | SM-D-875954 |
| Cable Interface | SM-D-876258 |
| Multi Vdc | SM-D-884567 |
| AC Input | SM-D-884579 |
| 180/180 Vdc | SM-D-884602 |

* This drawing contains automatic test equipment (ATE) acceptance procedures and programs.

3.13 Operational test. (See 4.7.7) Each fabricated and tested unit shall be subjected to a final operational test in accordance with 4.7.7, after all other inspections and tests are completed, to determine that the unit is operating prior to packaging. The operational test of group A satisfies this requirement for those units for which that is the last inspection or test.

3.14 Systems safety engineering. (See 4.13)

3.14.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirements 1 and 8, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual examination and inspection.

3.14.2 Rounding exposed edges. Exposed edges and corners shall be rounded sufficiently to preclude puncture or laceration hazards.

3.14.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes/spark gaps, lenses).

3.15 Nuclear survivability. (See 4.16) The equipment shall satisfy the nuclear survivability requirements stated in Section 3 of CR-CX-0173-001 with the exception of paragraphs 3.3.1.1 (BLAST) and 3.3.1.2 (THERMAL).

Source: https://assist.dla.mil - Downloaded: 2026-04-26T09:59Z
Check the source to verify that this is the current version before use.

3.16 Workmanship. (See 4.13) The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and with the applicable portions of MIL-P-11268 as follows:

a. Adhesives

b. Cables, cords and wires, electrical

c. Wire (harnessing)

d. Plastic material and parts

e. Rubber

f. Wiring and cabling

g. Connectors (electrical)

h. Printed wiring assemblies

i. Processes

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3) Does not include packaging.

b. Inspections covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections.

(1) Quality conformance inspections of equipment before packaging. (See 4.5)

(2) Packaging inspection. (See 4.14)

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspections shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish a minimum of 20 first article units of the complete equipment.The first article quantity shall be as specified in the contract or purchase order.

4.3.2 First article inspection. The first article inspections shall consist of inspections specified in table I and shall be performed in the order specified therein except that all hardwired and plug-in assemblies shall undergo inprocess group A inspection prior to burn-in.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

4.3.3 First article inspection data. The first article inspection test plan and test report(s) shall be as required in the contract or purchase order.

TABLE I. First article inspections.

| Inspection | Reqt Para | Test Para | Unit 1 2 | 3 4 5 | 6 7 8 9 | 10 11 | 12-19 | 20+ |
|---|---|---|---|---|---|---|---|---|
| 1. Inspections by Subsidiary Documents | N/A | 4.4 | Inspections to be performed on all units. | | | | | |
| 2. Burn-In | 3.6 | 4.6 | Inspections to be performed on all units. | | | | | |
| 3. Group A Inspection | (See Table II) | | Inspections to be performed on all units. | | | | | |
| 4. Group B Inspection | (See Table III) | | Inspections to be performed on all units. | | | | | |
| 5. Group C Inspection | | | | | | | | |
| Subgroup 1. | | | | | | | | |
| High Temperature | 3.8.1.1 | 4.8.1.1 | 1 | | | | | |
| Low Temperature | 3.8.1.2 | 4.8.1.2 | 2 | | | | | |
| Subgroup 2. | | | | | | | | |
| EMI | 3.9 | 4.9 | | 1 | | | | |
| Vibration/Shock | 3.8.7 | 4.8.7 | | 1 | | | | |
| Subgroup 3. | | | | | | | | |
| Humidity (Note C) | 3.8.2 | 4.8.2 | | | 1 | | | |
| Subgroup 4. | | | | | | | | |
| Altitude | 3.8.3 | 4.8.3 | | | 1 | | | |
| Dust (Note C) | 3.8.4 | 4.8.4 | | | 2 | | | |
| Salt Fog (Note C) | 3.8.5 | 4.8.5 | | | 3 | | | |
| Fungus (Note C) | 3.8.6 | 4.8.6 | | | 1 | | | |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspections. (continued)

| Inspection | Reqt Para | Test Para | Unit 1 2 3 4 5 6 7 8 9 10 11 12-19 20+ |
|---|---|---|---|
| Subgroup 5. | | | |
| Line-to-Ground Voltage | 3.7.1.2 | 4.7.1.2 | 1 ** |
| Clock/Data Phasing | 3.7.1.3 | 4.7.1.3 | 2 |
| Offset | 3.7.1.4 | 4.7.1.4 | 3 ** |
| Signal Balance | 3.7.1.5 | 4.7.1.5 | 4 |
| Slew Rate | 3.7.1.6 | 4.7.1.6 | 5 |
| Input Load Impedance | 3.7.1.8 | 4.7.1.8 | 6 |
| Clock Asymmetry | 3.7.1.9 | 4.7.1.9 | 7 |
| Output Rise and Fall Time | 3.7.2.2 | 4.7.2.2 | 8 |
| Output Impedance | 3.7.2.3 | 4.7.2.3 | 9 |
| Input Impedance | 3.7.2.4 | 4.7.2.4 | 10 |
| Input-Output Impedance | 3.7.3.1 | 4.7.3.1 | 11 |
| Frequency Response | 3.7.3.3 | 4.7.3.3 | 12 ** |
| Signal-to-Noise Ratio | 3.7.3.4 | 4.7.3.4 | 13 ** |
| Prime Power Interface | 3.7.5.1 | 4.7.5.1 | 14 |
| Power feed current limits | 3.7.5.3 | 4.7.5.3 | 15 |
| Current Protection | 3.7.5.4 | 4.7.5.4 | 16 ** |
| Full Duplex Signal Detection and Dynamic Range | 3.7.8 | 4.7.8 | 17 |
| Dimensional Interchangeability | 3.11 | 4.11 | 18 |
| 6. Group D Inspections Reliability | 3.10 | 4.10 | 8 units |

** One-time engineering test to be covered by separate engineering report. Not part of the first article test report.

NOTES:

a. The inspections 1 through 4, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

b. The numbers in the unit columns in the table covered by inspections 5 and 6 specify the order of inspections for the indicated unit.

c. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

d. All first article units shall be subjected to group A and group B inspection.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

| | Where required |
|---|---|
| Printed wiring | 3.2.2 |
| Semiconductor devices and integrated circuits | 3.2.3 |
| Parts, materials, and processes | 3.3 |
| Finish | 3.4 |
| Marking | 3.5 |

4.5 Quality conformance inspections of equipment before packaging. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.5.1 Group A inspection. Each unit on contract or purchase order and each subassembly and assembly identified in table II shall be inspected for conformance to the requirements as specified in table II.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in the order listed in table II. The order of inspection of the subassemblies, assemblies, and units (and equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in table II shall be subjected to a 100% visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in table II.

b. Only for spares that are not installed into end equipment the following shall be done: After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which passed the 100% inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105 using general inspection levels and AQLs indicated in table II (Quality Assurance Verification). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Assemblies and subassemblies which have passed the 100% visual/mechanical and electrical inspections of step a above shall be assembled into the end equipments, which shall then be subjected by the contractor to the burn-in test requirements of 4.6 and an end equipment 100% visual/mechanical and electrical inspection in accordance with table II in that order.

d. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from units which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using general inspection levels and AQLs indicated in table II. Group A electrical inspection shall be performed in the order shown in table II.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE II. Group A inspection.

| Inspection | Reqt Para | Insp Para | AQL Major | AQL Minor |
|---|---|---|---|---|
| VISUAL-MECHANICAL | | | | |
| Cable Interface CCA | 3.16 | 4.13 | 2.5% | 4.0% |
| Cable Driver CCA | 3.16 | 4.13 | 2.5% | 4.0% |
| BITE 1 CCA | 3.16 | 4.13 | 2.5% | 4.0% |
| Multi-Vdc CCA | 3.16 | 4.13 | 2.5% | 4.0% |
| AC Input CCA | 3.16 | 4.13 | 2.5% | 4.0% |
| 180/180 CCA | 3.16 | 4.13 | 2.5% | 4.0% |
| Case and covers | 3.16 | 4.13 | 2.5% | 4.0% |
| ELECTRICAL PERFORMANCE (Including Spare Assemblies) | | | | |
| Cable Interface CCA | 3.12 | 4.12 | 1.0% | * |
| Cable Driver CCA | 3.12 | 4.12 | 1.0% | * |
| BITE 1 CCA | 3.12 | 4.12 | 1.0% | * |
| Multi Vdc CCA | 3.12 | 4.12 | 1.0% | * |
| AC Input CCA | 3.12 | 4.12 | 1.0% | * |
| 180/180 CCA | 3.12 | 4.12 | 1.0% | * |
| ELECTRICAL | | | | |
| BITE and Alarms | 3.7.6 | 4.7.6 | 1.0% | * |
| Operational Tests | 3.7.7 | 4.7.7 | 1.0% | * |

* All electrical defects are major.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table III and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The group B AQL shall be as shown in table III and the inspection level shall be S-4.

4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in the order listed in table III.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE III. Group B inspections.

| Inspection | Reqt Para | Insp Para | AQL |
|---|---|---|---|
| Line-to-Line Voltage Levels | 3.7.1.1 | 4.7.1.1 | 4.0% |
| Receiver Dynamic Range | 3.7.1.7 | 4.7.1.7 | 4.0% |
| Output Voltage Levels | 3.7.2.1 | 4.7.2.1 | 4.0% |
| Receiver Sensitivity | 3.7.2.5 | 4.7.2.5 | 4.0% |
| Orderwire Output Level | 3.7.3.2 | 4.7.3.2 | 4.0% |
| Output Power Level | 3.7.4.1 | 4.7.4.1 | 4.0% |
| Power Feed Output Over-Voltage | 3.7.5.2 | 4.7.5.2 | 4.0% |

4.5.3 Group C inspection. Group C inspection shall be performed on units from lots that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for group C inspections.

4.5.3.1.1 Subgroup 1. For this subgroup, one unit shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every third month. The unit shall be subjected to high and low temperature in that order.

4.5.3.1.2 Subgroup 2. For this subgroup, two units shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every fourth month, and one unit shall be selected every eighth month. The unit selected every fourth month shall be subjected to vibration and shock testing in that order. The unit selected every eighth month shall be subjected to EMI testing.

4.5.3.1.3 Subgroup 3. For this subgroup, one unit shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every fifth month. The unit shall be subjected to humidity testing.

4.5.3.1.4 Subgroup 4. For this subgroup, two units shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every twelfth month. One unit shall be subjected to altitude, dust, and salt-fog testing, and one unit shall be subjected to fungus testing.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

4.5.3.1.5 Subgroup 5. For this subgroup, one unit shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every fourth month. The unit shall be subjected to the electrical tests in the order shown in table IV.

TABLE IV. Group C inspections.

| Inspection | Reqt Para | Insp Para |
|---|---|---|
| Subgroup 1 | | |
| High Temperature | 3.8.1.1 | 4.8.1.1 |
| Low Temperature | 3.8.1.2 | 4.8.1.2 |
| Subgroup 2 | | |
| Vibration | 3.8.7 | 4.8.7.1 |
| Shock | 3.8.7 | 4.8.7.2 |
| Functional | | 4.8.7.2.1 |
| Bench Handling | | 4.8.7.2.2 |
| EMI | 3.9 | 4.9 |
| Subgroup 3 | | |
| Humidity | 3.8.2 | 4.8.2 (Note 1) |
| Subgroup 4 | | |
| Low Pressure (Altitude) | 3.8.3 | 4.8.3 |
| Dust | 3.8.4 | 4.8.4 (Note 1) |
| Salt Fog | 3.8.5 | 4.8.5 (Note 1) |
| Fungus | 3.8.6 | 4.8.6 (Note 1) |
| Subgroup 5 | | |
| Dimensional Interchangeability | 3.11 | 4.11 |
| Clock/Data Phasing | 3.7.1.3 | 4.7.1.3 |
| Signal Balance | 3.7.1.5 | 4.7.1.5 |
| Slew Rate | 3.7.1.6 | 4.7.1.6 |
| Input Load Impedance | 3.7.1.8 | 4.7.1.8 |
| Clock Asymmetry | 3.7.1.9 | 4.7.1.9 |
| Output Rise and Fall Time | 3.7.2.2 | 4.7.2.2 |
| Output Impedance | 3.7.2.3 | 4.7.2.3 |
| Input Impedance | 3.7.2.4 | 4.7.2.4 |
| Input-Output Impedance | 3.7.3.1 | 4.7.3.1 |
| Prime Power Interface | 3.7.5.1 | 4.7.5.1 |
| Power Feed Current Limits | 3.7.5.3 | 4.7.5.3 |
| Full Duplex Signal Detection and Dynamic Range | 3.7.8 | 4.7.8 |

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

NOTE 1: The equipment shall be throughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

4.5.3.2 Order of inspection within group C. Group C inspections shall be performed in the order specified in 4.5.3.1.

4.5.3.3 Group C failures. Actions required relative to group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass group A inspection only.

4.5.5 Group D inspection. This inspection shall consist of the test specified in table V and shall be performed on units from lots which have been subjected to and have met group A and group B inspections. Test conditions, methods, and procedures shall be in accordance with 4.10.

TABLE V. Group D inspection.

| Inspection | Reqt para | Insp para |
|---|---|---|
| Reliability | 3.10 | 4.10 |

4.5.5.1 Sampling for group D inspection. Eight units shall be selected from the first month's production and subjected to the test of 4.10. For subsequent group D inspections, a maximum of eight units shall be selected at random from every fourth month's production.

4.5.5.2 Group D failures. Actions required relative to group D failures shall be as specified in the contract or purchase order.

4.5.5.3 Reinspection of conforming group D sample unit. Unless otherwise specified, sample units which have been subjected to and have passed group D inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass group A inspection only.

4.6 Electrical preconditioning (burn-in). (See 3.6) During electrical preconditioning, the equipment shall be connected per figure 1. Test conditions of 4.10.3 shall apply except that the

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

duty cycle shall be 100%. If a failure occurs during this preconditioning, the failure shall be recorded, the failed component(s) replaced, and the preconditioning cycle reinstituted. Failure data obtained shall be submitted with the lot containing the unit in which the failure occurred. The requirements of 3.6 shall be met.

a. Connect the equipment as shown in figure 1.

b. Set the Group Simulator as follows:

| | |
|---|---|
| Group Rate | 3 |
| Timing Mode | INT |
| Transmit Format | No Frame |
| Type | 2 |
| Channel Rate | 32 kb/s |
| Data Select | Random |

c. Set the unit under test (UUT) as follows:

| | |
|---|---|
| Group Rate | 3 |
| TRF Switches | ON |
| Remote Power Out (switches A and B) | ON |

d. Turn power on, on all UUT's. Run the units for 48 hours, using BITE to indicate failure. The last 24 hours of burn-in shall be failure free.

e. Provisions shall be made to latch all S, SA and LS alarms.

f. After the completion of burn-in, perform the operational test procedure specified in 4.7.7.

4.7 Electrical. (See 3.7) Unless otherwise specified, all tests shall be performed at 115 Vac ±5% from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperature, pressure, and humidity per MIL-STD-810D unless otherwise specified. The electrical test equipment listed in table VI or their equivalent will be utilized during testing and will be calibrated in accordance with MIL-STD-45662. The standard test equipment shall be provided by the contractor. Government technical representative approval is required prior to use of proposed substitute items. Unless otherwise specified, special test equipment (see table VII) shall be provided by the contractor subject to approval by the Government. The Government shall be responsible for supplying the Government Furnished Equipment (GFE) required for test. (See table VIII). Table IX identifies the external interface connections used in the test figures. Unless otherwise noted, all resistors in the figures shall be 0.5 watt.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE VI. Standard test equipment.

| Description | Mfg./model no. (or equivalent) |
|---|---|
| Oscilloscope | Tektronix 7904 |
| Wattmeter | Weston 432 |
| Voltmeter, Digital | HP 3440A |
| Voltmeter, AC | HP 403 |
| Supply, AC Variable Frequency | Elgar 1001 |
| Audio Oscillator | HP 200CD |
| Attenuator, 50-ohm | Kay 432 |
| Milliammeter DC | Weston 931- 5, 20, 100 and 500 milliamperes |
| True RMS Voltmeter | Ballantine Model 323 |
| Transmission Measurement Set | HP 3555B |

Table VII. Special test equipment.

| Equipment name | Drawing no. | Description |
|---|---|---|
| Group Simulator | A-3051667 | The Group Simulator simulates the group data, orderwire and telemetry signals. It generates a long or short pseudo-random sequence. The three signals can be generated simultaneously or in any combination. A TTL error output signal is provided for counting errors externally. |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE VIII. Government furnished equipment (GFE).

| Equipment | Description |
|---|---|
| CX-11230()/G Cable | Twin coax cable supplied as GFE for test. 1/4-mile reels. |
| Modem, Cable Driver MD-1024/G | The High Speed Cable Driver Modem provides high voltage cable power at 90 ma for repeated cable transmission. |
| Alternate (for Group Simulator): Generator, Digital Data SG-1139/G | As an alternate for the DSVT Simulator, the Digital Data Generator provides pseudo-random data patterns of 72 to 4,608 kb/s balanced signals and 16/32 kb/s conditioned diphase signals. It provides correlation of digital data to perform bit-error measurements. |

TABLE IX. MD-1025( )/G external interface connections.

| Connector | Function | Pin | | | Signal Type |
|---|---|---|---|---|---|
| | | HI | LO | GND | |
| LINE SIGNAL IN 1 | Line data in | - | - | - | DIPHASE |
| LINE SIGNAL IN 2 | Line data in | - | - | - | DIPHASE |
| LINE SIGNAL OUT 1 | Line data out | - | - | - | DIPHASE |
| LINE SIGNAL OUT 2 | Line data out | - | - | - | DIPHASE |
| Group 1 | Group Data in<br>Group Tim in | B<br>D | A<br>C | M<br>N | NRZ<br>SQ WV |
| Group 2 | Group Data out<br>Group Tim out | F<br>H | E<br>G | P<br>R | NRZ<br>SQ WV |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE IX. MD-1025( )/G interface connections-continued.

| Connector | Function | Pin HI | Pin LO | Pin GND | Signal Type |
|---|---|---|---|---|---|
| ORDERWIRE and ALARM | Alarms S | B | A | | Alm= Close contact |
| | Alarms SA | D | C | | Alm= Close contact |
| | Alarms LS | F | E | | Alm=Close content |
| | AVOW #1 in | H | G | T | Analog |
| | AVOW #1 out | K | J | | Analog |
| | AVOW #2 in | M | L | | Analog |
| | AVOW #2 out | P | N | | Analog |
| POWER | AC input | A | C | B | |

4.7.1 Balanced equipment-side signal characteristics. (See 3.7.1)

4.7.1.1 Line-to-line voltage levels. (See 3.7.1.1)

a. Connect the equipment as shown in figure 2.

b. Set the UUT as follows:

| | |
|---|---|
| Group Rate | 3 |
| TRF Switches | ON |
| Remote Power Out Switches | OFF |

c. Set the Group Simulator as follows:

| | |
|---|---|
| Group Rate | 144 kb/s |
| Timing Mode | MASTER |
| Transmit Format | No Frame |
| Type | 2 |
| Channel Rate | 32 kb/s |
| Data Select | RANDOM |



Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

d. Measure the Group and Timing Out signals. Measure the positive and negative voltage levels line-to-line.

e. Verify that the requirements of 3.7.1.1 are met.

4.7.1.2 Line-to-ground voltage. (See 3.7.1.2) Verify the line-to-ground voltage using the test configuration in figure 3 by measuring from each line to ground. Verify that the requirements of 3.7.1.2 are met.

4.7.1.3 Clock/data phasing. (See 3.7.1.3)

a. Connect the equipment as shown in figure 3.

b. Set the UUT as follows:

| | |
|---|---|
| (1) TRF Switches | ON |
| (2) Group Rate Switches | 3 |
| (3) Remote PWR OUT Switches | OFF |

c. Set the Group Simulator as follows:

| | |
|---|---|
| (1) Group Rate | 144 kb/s |
| (2) Timing Mode | MASTER |
| (3) Transmit Format | No frame |
| (4) Data Select | RANDOM |

d. Measure the delay between the zero crossings on both the data rise and fall slopes to the zero crossing of the timing rise slopes as indicated in figure 4.

e. Verify that the requirements of 3.7.1.3 are met.

4.7.1.4 Offset voltage. (See 3.7.1.4)

a. Connect the equipment as shown in figure 5, operating as specified in 4.7.1.3.

b. Using the oscilloscope, measure the dc offset voltage at the Equipment Group 1 Timing and Data outputs.

c. Measure the offset voltage tap of the 100-ohm load to ground.

d. Repeat steps b and c for Equipment Group 2.

e. Verify that the requirements of 3.7.1.4 are met.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.1.5 Signal balance. (See 3.7.1.5)

a. Connect the equipment as shown in figure 2.

b. Compare the difference between the positive and negative values of the Group 1 and 2 Timing Out signals.

c. Repeat step b for the Data Out signals.

d. Verify that the difference between the positive and negative values are within 10%. Difference is divided by the larger of the two.

4.7.1.6 Slew rate. (See 3.7.1.6)

a. Connect the equipment as shown in figure 6. The UUT and Group Simulator settings are the same as in 4.7.1.3.

NOTE: All measurements shall be made through 100 feet of RG-108 cable.

b. With the UUT at Group Rate 3, measure the time between the +40 mV and -40 mV points around the zero voltage level of the rise and fall slopes of the Group Data and Timing Out signals as shown in figure 7.

c. Verify that the time measured in step b is not more than 10 ns for each signal.

4.7.1.7 Receiver dynamic range. (See 3.7.1.7)

a. Connect the equipment as shown in figure 8. Dual variable attenuators will be connected in the signal path during the test.

b. The Group Simulator and UUT switch settings are the same as specified in 4.7.1.1.

c. Set the UUT for Group Rate 3 operation. Connect the attenuators in the Group 1 Data and Timing signal paths. Using the oscilloscope, adjust each attenuator to give a $\pm$50 mV peak signal referenced to ground.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

d. Verify that there are no errors for a period of 30 seconds.

e. Repeat steps c and d for Group 2 input.

4.7.1.8 Input load impedance. (See 3.7.1.8)

a. Connect the equipment as shown in figure 9.

b. Adjust the oscillator to 1 kHz and an output voltage of 1.00 volt RMS measured at V1. Measure the V2 voltage at the Group 1 Timing input.

c. Calculate the impedance using Z in = 100 $(V_2/V_1)$.

d. Repeat steps b and c for the Group 1 Data input and for the Group 2 Timing and Data inputs.

e. Verify that the requirement of 3.7.1.8 is met.

4.7.1.9 Clock asymmetry. (See 3.7.1.9)

a. Connect the equipment as shown in figure 2.

b. Measure the Group 1 Timing output clock waveform on the oscilloscope.

c. Verify that the positive half of the Group 1 clock waveform is equal to half the clock period within $\pm$120 ns measured at 50% point as shown in figure 10.

d. Repeat steps b and c for the group 2 Timing Output.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.2 Cable-side group signal characteristics. (See 3.7.2)

4.7.2.1 Output voltage levels. (See 3.7.2.1)

a. Connect the equipment as shown in figure 11.

b. The Group Simulator and UUT switch settings are the same as specified in 4.7.1.1.

c. Verify, using the oscilloscope, that the requirements of 3.7.2.1 are met.

d. Repeat steps b and c for Group 2 input and Line Signal 2 output.

4.7.2.2 Output rise and fall time. (See 3.7.2.2)

a. Connect the equipment as shown in figure 11.

b. Set the UUT switches as specified in 4.7.1.1.

c. Measure the group output rise and fall time. (See figure 12)

d. Verify that the requirement of 3.7.2.2. is met.

4.7.2.3 Output impedance. (See 3.7.2.3)

a. Connect the equipment as shown in figure 13 to the Line Signal OUT 1.

NOTE: Check that the UUT REMOTE POWER OUT switches A and B are in the OFF position.

b. Measure the signal output voltage with S1 open (V1) and with S1 closed (V2).

c. Calculate the output impedance using the formula:

$$Z_{out} = 57.6\left(\frac{V1}{V2} - 1\right) \text{ ohms}$$

d. Repeat a through c for Line Signal Out 2.

e. Verify that the requirements of 3.7.2.3 are met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.2.4 Input impedance. (See 3.7.2.4)

a. Connect the equipment as shown in figure 14. Connect the oscillator output to the Line Signal Input 1. Check that the UUT REMOTE POWER OUT switches A and B are in the OFF position.

b. Adjust the oscillator to 300 Hz and set the output level to 1.00 volt rms measured at V1.

c. Measure the V2 voltage. Calculate the impedance using Z in = 57.6 (V2/V1).

d. Adjust the oscillator to 300 kHz and repeat steps b and c.

e. Connect the oscillator output to the Line Signal Input 2.

f. Repeat steps b through d for Line Signal Input.

g. Disconnect the oscillator from the Line Signal Input.

4.7.2.5 Receiver sensitivity. (See 3.7.2.5)

a. Connect the equipment as shown in figure 15 with the Group Simulator and UUT switch settings as specified in 4.7.1.1.

b. Verify that there are no error counts for a period of 30 seconds.

c. Connect the Group Simulator to group 2 and repeat b.

4.7.3 Equipment-side maintenance orderwire signal characteristics. (See 3.7.3)

4.7.3.1 Orderwire input and output impedance. (See 3.7.3.1)

a. Connect the equipment as shown in figure 16. Connect the oscillator output to the Analog Voice Orderwire (AVOW)1.

b. Adjust the oscillator to 1 kHz and set the output level to 1.00 volt rms at $V_1$.

c. Measure the V2 voltage. Calculate the impedance using $Z \text{ in} = 604 \frac{V2}{V1}$.

d. Repeat steps b and c with the oscillator frequency set at 300 Hz and 3400 Hz.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

e. Repeat steps b through d with the oscillator connected to AVOW 1.

f. Repeat steps b through e for the AVOW 2 input and output.

g. Verify that the requirements of 3.7.3.1 are met.

4.7.3.2 Orderwire output level. (See 3.7.3.2)

a. Connect the equipment as shown in figure 17. Set the UUT TRF switches to the OFF position.

b. Connect the Audio Oscillator to the Line Signal 1 input and adjust the oscillator amplitude to 0.411 Vrms (-5.5dBm) at 1000 Hz.

c. Connect the rms Multimeter to the AVOW 1 output.

d. Verify that the requirements of 3.7.3.2 are met.

e. Adjust the oscillator amplitude to 1.227 Vrms (+4 dBm)and repeat step d.

f. Repeat steps b through e for the AVOW 2 output.

4.7.3.3 Frequency response. (See 3.7.3.3)

a. Connect equipment as shown in figure 18.

b. Set the oscillator frequency to 1 kHz and adjust the level for +2 dBm and then connect to the AVOW 1 input.

c. Connect I/O Pin 29 to ground on CCA A4.

d. Measure the AVOW 1 output signal level.

e. While maintaining the level at +2 dBm, adjust the oscillator frequency to 300 Hz, 400 Hz, 3000 Hz, and 3400 Hz. At each frequency measure the output power in dB relative to the power at 1 kHz.

f. Repeat steps b through d for AVOW 2 with I/O pin 50 grounded.

g. Verify that the loss versus frequency characteristics of the orderwire groups are within the limits specified in 3.7.3.3.

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.3.4 Signal-to-noise ratio. (See 3.7.3.4)

a. Connect the equipment as shown in figure 19 with Group Simulator connected to Group 1.

b. Set the UUT operation for Group Rate 1 (72 kb/s), Traffic switches ON, Power Feed switches OFF.

c. Set the Group Simulator for Internal Timing, Group Rate 1, No Frame, and Random Data. Connect pin 29 to ground on CCA A4.

d. Measure the AVOW 1 output level in dB, C-weighted. This is V noise with AVOW 1 input disconnected.

e. Set the audio oscillator frequency to 1 kHz and adjust the output level to 2 dBm with AVOW 1 input connected.

f. Measure the AVOW 1 output level in dB. This is V Signal.

g. Move the Group Simulator to Group 2.

h. Repeat steps d, e and f for AVOW 2 with I/O pin 50 grounded on CCA A4.

i. Calculate the signal-to-noise ratio for AVOW 1 and AVOW 2 using the formula SNR =V Signal - V Noise.

j. Verify that the requirements of 3.7.3.4 are met.

k. Set the Group Simulator for Group Rate 3 and connect the Group Simulator to Group 1.

l. Repeat steps d through j.

4.7.4 Cable-side maintenance orderwire signal characteristics.

4.7.4.1 Output power level. (See 3.7.4.1)

a. Connect the equipment as shown in figure 20. Set the UUT TRF switches to the OFF position.

b. Connect the audio oscillator to the AVOW 1 input and adjust the oscillator amplitude to -4.0 dBm at 1000 Hz.

c. Connect the rms multimeter to the Line Signal 1 output.

d. Verify that the OW output power level is +5 dBm ±1 dB.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

e. Connect the audio oscillator to the AVOW 2 input and adjust the oscillator amplitude to -4.0 dBm at 1000 Hz.

f. Connect the rms multimeter to the Line Signal 2 output.

g. Measure the OW output power level. Verify it measures +5 dBm ±1 dB.

4.7.5 Power interface. (See 3.7.5)

4.7.5.1 Prime power. (See 3.7.5.1)

a. Connect the equipment as shown in figure 21. The Group Simulator switch setting shall be as specified in 4.7.1.3.

b. On the UUT, set the REMOTE POWER OUT switches A and B ON.

c. Measure the input power and monitor the Group Data BER over a 30-second period for each of the following combinations of input voltage and frequency:

| | |
|---|---|
| 115 Vac | 60 Hz |
| 103.5 Vac | 47.5 Hz and 420 Hz |
| 126.5 Vac | 47.5 Hz and 420 Hz |

4.7.5.2 Power feed output over-voltage. (See 3.7.5.2)

a. Connect the equipment as shown in figure 22 except that a storage oscilloscope is used in place of the digital voltmeter.

b. Connect the Line Signal Out 1 to a 600-ohm load.

c. Disconnect the output from the load by switching to V (open) switch position and observe the maximum open circuit voltage as recorded on the storage oscilloscope.

d. Verify that the requirements of 3.7.5.2 are met.

4.7.5.3 Power feed current limits. (See 3.7.5.3)

a. Connect the equipment as shown in figure 22.

b. Connect the Line Signal Out 1 to a 12.5-kohm load.

c. Turn on the UUT REMOTE POWER OUT switch A and verify that the power supply remains ON.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check this source to verify that this is the current version before use.

d. Switch the Line Signal Out 1 to a 33-kohm load.

e. Verify that the power shuts down.

f. Repeat b through e substituting a 402-ohm load in d.

g. Repeat b through f for Line Signal Out 2.

h. Measure and verify that the following requirements are met:

| Measurement | Requirement |
|---|---|
| (1) Minimum load current (Step b) | Power supply remains ON and output voltage is between 170 and 190 Vdc |
| (2) Undercurrent shutdown (Step d) | Power supply shuts down when current drops below 10 ma |
| (3) Overcurrent protection (Step f) | Power supply shuts down when current is 420 ma or greater |

4.7.5.4 Current protection. (See 3.7.5.4)

a. Connect the equipment as shown in figure 23.

b. With the UUT AC Power turned OFF, connect the equipment Line Signal out to the UUT Line Signal Input 1.

c. Turn on the MD-1024 Power Feed for 1 minute.

d. Verify there is no damage to the UUT.

e. Repeat b through e for the Line Signal 1 Output and for the Line Signal 2 Input and Output.

f. Perform the following operational tests on the UUT:

(1) Group error check in accordance with 4.7.7.1.

(2) Alarm monitor check in accordance with 4.7.7.2.

(3) DC voltage check in accordance with 4.7.7.3.

(4) Maintenance orderwire check in accordance with 4.7.7.4.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.6 Built-in test equipment (BITE) alarm monitor check. (See 3.7.6.)

a. Connect the equipment as shown in figure 24 with the 600-ohm power feed resistors disconnected.

b. Verify that the TRF switches on CCAs A5 and A6 are set to ON, the Group Rate switches to 3 and the REMOTE POWER OUT switches OFF.

c. Set the Group Simulator Timing to Group Rate 3, No Frame, Random Data, Internal Timing.

d. Verify that the following BITE conditions are met:

(1) S Alarm open

(2) SA Alarm open

(3) LS Alarm open

(4) Summary Alarm Lamp OFF

(5) Power Supply Alarm Lamp OFF

(6) All CCA LEDs OFF

(7) Loss of Input Lamp OFF

e. Turn REMOTE POWER OUT switches A and B OFF.

f. Disconnect Line Signal 1 OUT and IN. Verify that the following BITE and Alarm conditions are met:

(1) S Alarm closed

(2) LS Alarm closed

(3) SA Alarm open

(4) Summary Alarm lamp illuminates

(5) LED of CCA A5 illuminates



Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

g. Set the UUT REMOTE POWER OUT switch A ON. Verify that the following BITE and Alarm conditions are met:

(1) S Alarm closed

(2) SA Alarm closed

(3) LS Alarm closed

(4) Summary Alarm Lamp illuminates

(5) Power Supply Alarm Lamp illuminates

(6) CCA A3 and A5 LED ON

h. Reconnect Line Signal 1.

i. On CCA A5 set the Group Rate switch to 1. Verify that the folloing BITE and Alarm conditions are met:

(1) S Alarm closed

(2) SA Alarm closed

(3) LS Alarm open

(4) Summary Alarm Lamp illuminates

(5) LED of CCA A5 illuminates

j. Set the Group Rate switch to 3 on CCA A5.

k. Repeat f through j using Line Signal 2, REMOTE POWER OUT B, Group Rate switch on CCA A6, and verifying the LED's on CCA's A3 and A6.

l. Press and hold the Lamp test switch on the UUT. Verify that the following BITE requirements are met:

(1) S Alarm closed

(2) SA Alarm closed

(3) LS Alarm closed

(4) Summary, Power Supply Alarm and LOI Lamps illuminate

(5) BITE LED's on CCA's illuminate during self test

m. Release test button and verify that the BITE indicators are the same as d above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

n. With UUT POWER ON, remove power at the source. Verify that the following Alarm conditions are met.

(1) S Alarm closed

(2) SA Alarm closed

(3) LS Alarm open

o. Turn UUT power switch OFF at the UUT. Verify that the SA Alarm opens.

4.7.7 Operational tests. (See 3.7.7)

4.7.7.1 Group error check. (See 3.7.7.1)

a. Connect the equipment as specified in 4.7.6.

b. Set the UUT TRF switches ON and GP Rate switches to 3.

c. Set the UUT REMOTE POWER OUT switches A and B ON.

d. Verify that there are no group errors for a period of 30 seconds.

4.7.7.2 Maintenance OW output check. (See 3.7.7.2)

a. With the 600-ohm power feed resistors disconnected, set the audio oscillator to -4 dBm at 1 kHz while connected to the UUT, and measure the AVOW 1 output level on the digital multimeter.

b. Repeat a for AVOW 2.

c. Verify that the requirements of 3.7.7.2 are met.

4.7.7.3 Output dc voltage check. (See 3.7.7.3) Measure the Line Signal 1 and Line Signal 2 dc output voltage using the digital multimeter with the 600-ohm power feed resistors connected. Verify that the requirements of 3.7.7.3 are met.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.7.4 Alarm monitor check. (See 3.7.7.4) Verify that the following BITE conditions are met:

a. S Alarm open

b. SA Alarm open

c. LS Alarm open

d. Summary Alarm Lamp OFF

e. Power Supply Alarm and Loss of Input Lamps OFF

f. All CCA LEDs OFF

4.7.8 Full duplex signal detection and dynamic range. (See 3.7.8)

a. Connect the equipment as shown in figure 25 for operation with Group 1 of the UUT.

b. Set the Group Simulator for Group Rate 3, Internal Timing, No Frame, and Random data.

c. Set the UUT TRF switches ON, Group Rate switches to 3 and Remote Power Out switches OFF.

d. Adjust the oscillator output level to -4 dBm at 1 kHz as measured at the AVOW 1 OW input.

e. Monitor the Group Simulator Error Output for 30 seconds. Verify that there are no errors.

f. Remove CX11230 cable and attenuators and loop Line Signal 1 to Line Signal 2. Repeat d and e.

g. Set the Group Simulator and the UUT for Group Rate 1.

h. Repeat d through f.

4.7.9 Environmental operational tests. (See 3.7.9) Perform operational tests as specified in 4.7.7.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.8 Service conditions tests. (See 3.8)

4.8.1 Temperature. (See 3.8.1)

4.8.1.1 High temperature. (See 3.8.1.1) The equipment shall be subjected to the test of MIL-STD-810D, Method 501.2 Procedure II for cycling exposure. In step 12, raise the internal chamber temperature to +63°C (+145°F). Maintain internal chamber temperature for 6 hours at +63°C (+145°F). Raise the internal chamber temperature to +71°C (+160°F) within a time period of one hour and maintain at that temperature for four additional hours. Lower the internal chamber temperature to +63°C (+145°F) within a time period of one hour. Repeat the above steps two additional times for a total of three 12-hour cycles. Step 14 shall be performed as follows: adjust the internal chamber temperature to +63°C (+145°F) and maintain this temperature until temperature stabilization of the test item is reached. Operate the test item until the item is stabilized.

4.8.1.2 Low temperature. (See 3.8.1.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 502.2, Procedure I. Lower the internal chamber temperature in step 2 to -57°C (-70°F) for a period of not less than two hours following stabilization of the equipment, or until 24 hours have elapsed, whichever occurs first. In step 5 of Procedure 1, perform Procedure II - Operation. (Para. II - 3.2 of Method 502.2). The chamber temperature in step 1 of Procedure II shall be adjusted to -31.7°C (-25°F).

4.8.2 Humidity. (See 3.8.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 507.2, Procedure III.

4.8.3 Low Pressure (Altitude). (See 3.8.3) The equipment shall be subjected to the test of MIL-STD-810D, Method 500.2, Procedure I. For step 2, adjust the chamber pressure to 141mm of Hg (5.56 inches of Hg or 40,000 feet above sea level), at a rate not to exceed 2,500 fpm. The equipment shall also be subjected to the test of MIL-STD-810D, Method 500.2, Procedure II. The chamber pressure in step 2 shall be adjusted to 523 mm of Hg (20.6 inches of Hg or 10,000 feet above sea level) at a rate not to exceed 2,500 fpm.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.8.4 Dust. (See 3.8.4) The equipment shall be subjected to the test of MIL-STD-810D, Method 510.2, Procedure I. In step 2 and step 5, the air velocity shall be 300 + 200 feet per minute. In step 5, the air temperature shall be adjusted to +63°C(+145°F).The equipment shall be reoriented periodically to expose all of its sides to the dust stream. The equipment shall be operated before step 1, during step 8 and during step 11.

4.8.5 Salt fog. (See 3.8.5) The equipment shall be subjected to the test of MIL-STD-810D, Method 509.2, Procedure I. In step 2, the 48-hour period shall be constant wetting. Delete step 3. Operation in step 4 shall take place after gentle wash in running water, not warmer than +38°(+100°F), and drying. There shall be no corrosion of finishes or metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish on metal surfaces. There shall be no clogging or binding of moving mechanical parts.

4.8.6 Fungus. (See 3.8.6) The equipment in the assembled-and-ready-for-delivery condition, or representative samples subject to approval of the procuring agency, shall be subjected to the test of MIL-STD-810D, Method 508.3. After 28 days, based on visual examination, the equipment or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than 6 unrelated minute colonies, with mycelial development in non-critical areas. Critical areas include: terminal spacing, CCA's etc. The equipment shall fail if it shows more than the growth specified above.

4.8.7 Vibration and shock. (See 3.8.7)

4.8.7.1 Vibration. (See 3.8.7) The equipment shall be subjected to the test of MIL-STD-810D, Method 514.3, Procedure I, Category 8 (Ground Mobile). The vibration level shall be in accordance with figures 514.3-7 through 514.3-9 of MIL-STD-810D. The axis designated on the schedules are the vehicle axis. The orientation of the equipment relative to the vehicle, therefore, must be considered for proper direction of vibration input. Test time shall be 4 hours in each of 3 mutually perpendicular directions for a total of 12 hours. Prior to, during (10 minutes of each hour), and following these tests, the equipment shall be operated in accordance with MIL-STD-810D general test performance guidance paragraph 4.5. The equipment shall not have any degradation in performance and shall not suffer any mechanical damage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.8.7.2 Shock. (See 3.8.7)

4.8.7.2.1 Functional. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure I. The shock pulse shall be 20g peak for a nominal duration of 11 msec in accordance with figure 516.3-4.

4.8.7.2.2 Bench handling. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure VI.

4.9 EMI tests. (See 3.9) The unit shall be tested for compliance with 3.9. Test methods and procedures shall be in accordance with MIL-STD-462, Notice 3, and a Government approved test plan. During susceptibility testing, the integral test facility plug-in subassembly shall be used to monitor operation of the unit.

4.9.1 Bonds and grounds. (See 3.9.1) Prior to performance of EMI testing, compliance with requirements of 3.9.1 shall be ascertained. Bonding resistance measurements shall be recorded and included in the EMI test report.

4.10 Reliability testing. (See 3.10)

4.10.1 First article reliability test. First article reliability testing shall be in accordance with Test Plan VIC, of MIL-STD-781C.

4.10.2 Quality conformance (Group D) reliability test. The reliability test for Group D inspection shall be in accordance with Test Plan VIC, of MIL-STD-781C.

4.10.3 Reliability test conditions. Reliability testing shall be accomplished subject to the following conditions:

a. The equipment shall be operated back-to-back.

b. Duty cycle: Apply prime power to equipment under test for 4 hours. Turn power OFF for 2 hours. Turn power ON for 16 hours. Turn power OFF for 2 hours. Repeat this sequence for each 24-hour cycle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

c. Chamber temperature shall be 40 ±5°C (95°F to 113°C).

d. Vibration shall be 2.2g ±10% peak acceleration value at any nonresonant frequency between 20 and 60 cps measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.

4.10.4 Reliability test method. The detailed test procedures and conditions for performing the tests outlined in 4.10.1 and 4.10.2 shall be in accordance with a contractor prepared, Government approved test plan.

4.10.5 Failure definition. A failure is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, or degradation below limits specified in 3.7.7.

4.10.6 Failure reporting and analysis. A failure reporting and analysis system shall be implemented per MIL-STD-781. Final determination of the failure relevancy and multiple and pattern failure categorization shall be the responsibility of the procuring activity.

4.11 Interchangeability. (See 3.11) The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.11. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of switches, sockets, and fasteners.

d. Location of panel-mounting screw holes on basic unit for mating with case.

e. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts or plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.12 Electrical subassembly tests. (See 3.12) Unit subassemblies, including spares, shall be tested in accordance with the test procedures specified on their respective test requirements specification drawings listed in 3.12.

4.13 Visual and mechanical inspection. (See 3.14 and 3.16) The equipment shall be examined for the defects listed in MIL-STD-252 to determine compliance with 3.16. An inspection shall be performed to verify compliance with those portions of 3.14 (Systems safety engineering) which can be determined visually.

4.13.1 System Safety Design Verification Check List. An inspection shall be performed, on one unit only, during first article testing, utlizing SEL Form 1183, to verify compliance with 3.14 (Systems safety engineering.)

4.14 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

4.14.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.14.2 Quality conformance inspection of packaging.

4.14.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.14.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.14.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table X. Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and AQL of 4.0% defective.

TABLE X. Packing inspection provisions.

| NO. | CHARACTERISTIC | METHOD OF INSPECTION |
|---|---|---|
| 101 | Intermediate container not as specified | VISUAL |
| 102 | Improper closure of intermediate container | VISUAL |
| 103 | Shipping containers not in accordance with specification | VISUAL |
| 104 | Excessive cube | VISUAL |
| 105 | Improper blocking and bracing | VISUAL |
| 106 | Closure not in accordance with specification | VISUAL |
| 107 | Weight and size exceed container limitation | WEIGHT & VISUAL |
| 108 | Strapping not in accordance with specification, incorrectly applied, omitted | VISUAL |
| 109 | Marking omitted, incorrect, or illegible | VISUAL |

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.15 Serial numbers. The equipment shall be inspected to verify that the serial numbers appear on each nomenclatured item and CCA.

4.16 Nuclear tests. Tests for nuclear survivability shall be performed on the equipment in accordance with Section 3 of CR-CX-0173-001.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-2073-2, tables I through VII coded as follows:

5.1.1 Level A. 3Q-1-00-00-NR-X-ED.

5.1.2 Level B. 2M-1-00-00-NR-X-ED.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The MD-1025/G is a shelter-mounted equipment providing cable interface for signal and power between TRI-TAC shelters and the exposed TD-1233/TTC and associated DSVT's via CX-11230 cable. It operates at 72 kb/s or 144 kb/s. It also provides transmission of maintenance orderwire.

6.1.1 Functional description. The equipment accepts non-return-to-zero (NRZ) equipment-side inputs and converts them to conditioned diphase for transmission over CX-11230 for distances up to 3.2 km (2 miles). Conditioned diphase signals received from the cable are converted to NRZ for output on the equipment side. The equipment provides sufficient power via the cable to operate the Multiplexer TD-1233/TTC and its associated Digital Secure Voice Terminals (DSVT's), KY-68. The equipment is designed for use in shelters and vans and is mounted in an equipment case. Each equipment case contains two modem modules. The two modules are functionally independent and failure of one module does not degrade the operation of the other module.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or B of preservation, and packing (see Section 5).

c. When first article rough handling tests are not required.

d. Quantity of first article inspection units required.

e. Place of final inspection.

f. Technical literature required.

g. Quantity of tools and running spare parts required.

h. Group C failure requirement.

i. MIL-Q-9858.

j. Length of cables (if any are required).

k. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

6.3 Environmental. Environmental pollution prevention measures are contained in the packing material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Final inspection. (See 4.5.3.3) Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.4.1 Initial shipments. (See 4.5.3.3) Initial shipments of end items will be contingent upon successful completion of all first article tests.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

6.5 Location of operational inspection. (See 4.7.7) It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

6.8 Government-furnished property. The contracting officer should arrange to furnish the property listed in 4.7.

6.9 Changes from previous issue. Asterisks (*) or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.10 Subject terms (key words) list.

| | |
|---|---|
| Balanced | Modem |
| Conditioned Diphase | Modem, Cable Driver |
| Digital Group Multiplex | Modem, Cable Driver, MD-1025(P)/G |
| Diphase | Orderwire |
| Group | Remote Loop Group Multiplexer/Cable Driver (RLGM/CD) |
| Maintenance Orderwire | |

CUSTODIAN:

ARMY - CR

PREPARING ACTIVITY:

ARMY - CR
Project No. 5820-A824

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

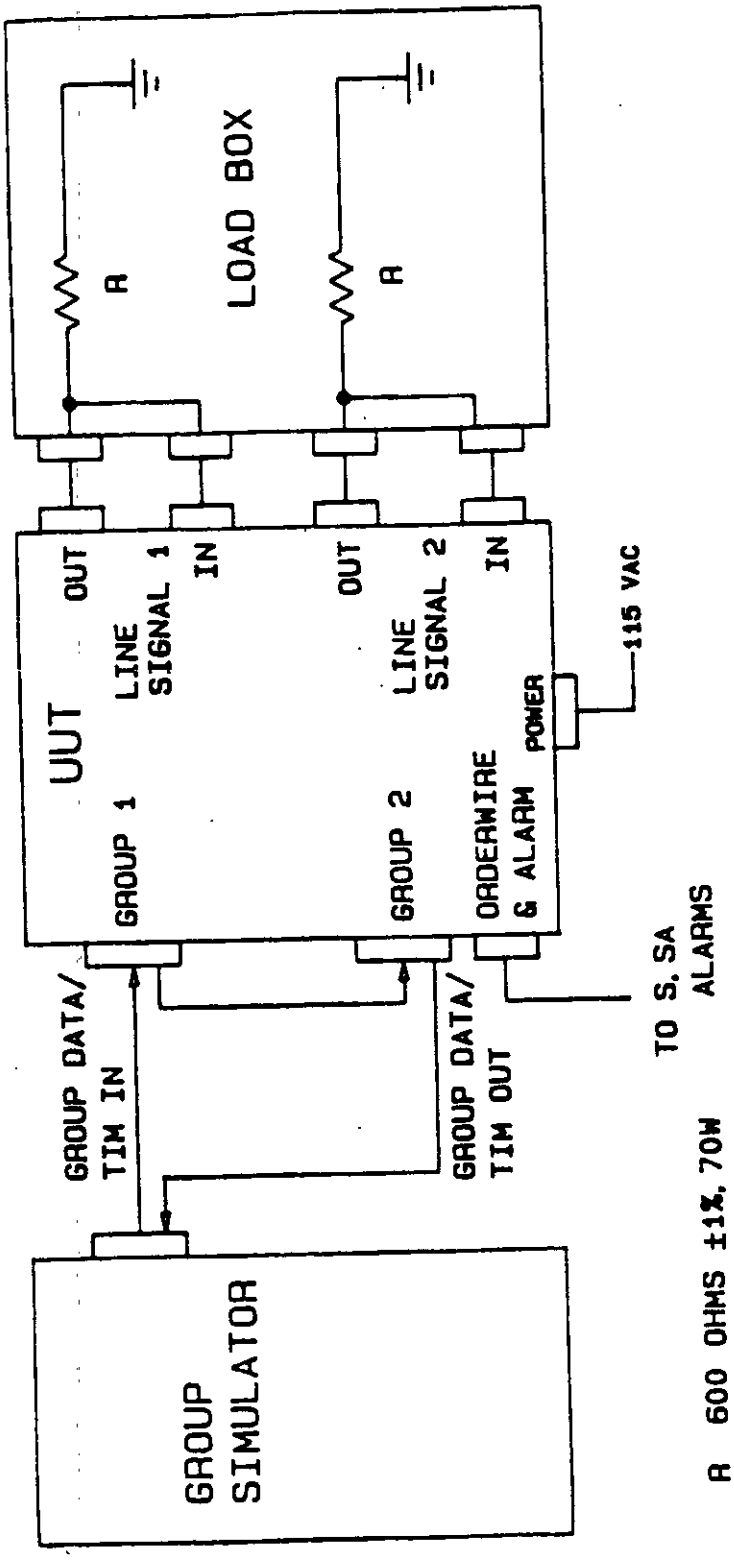



FIGURE 1. Burn-in.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

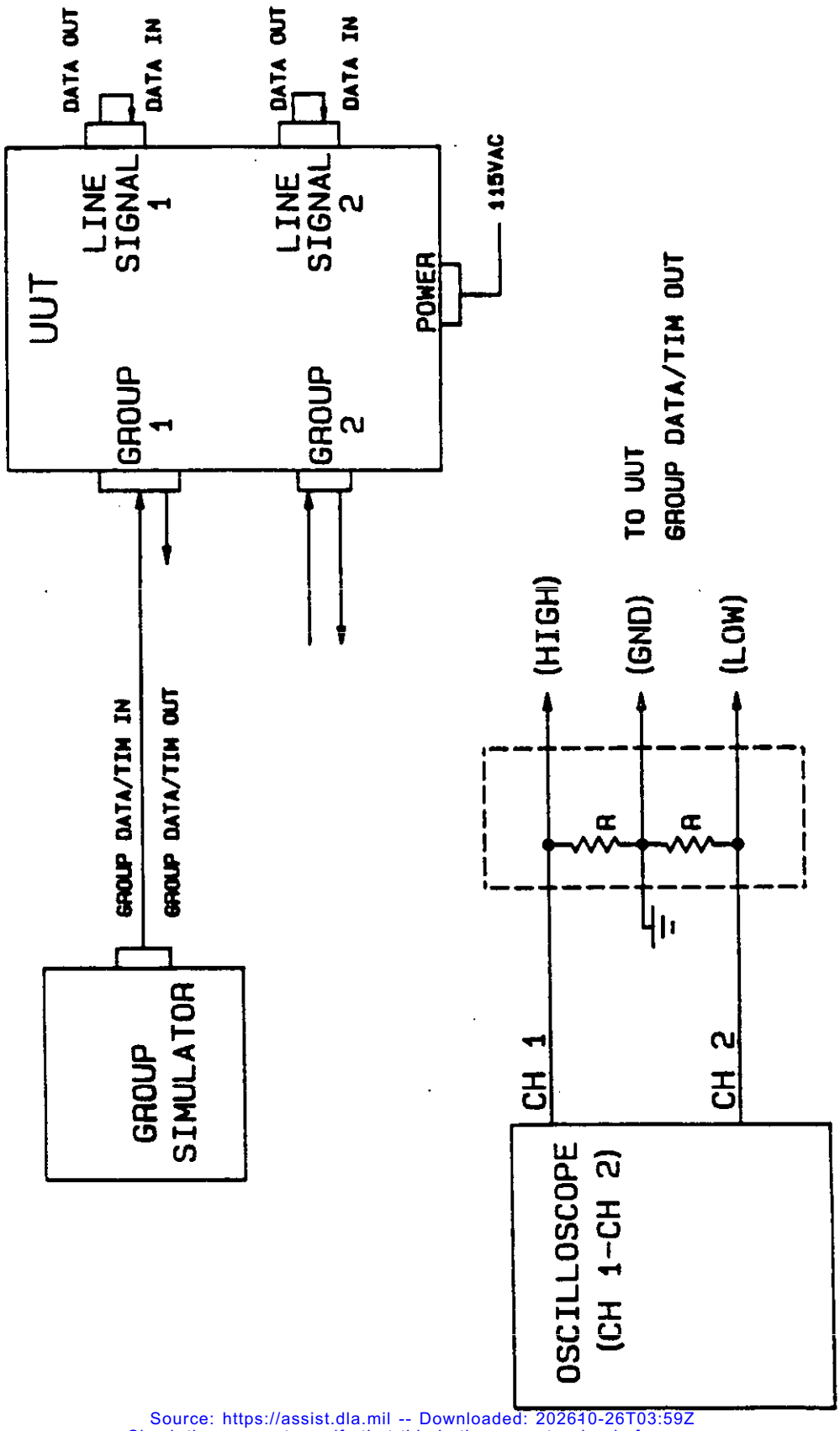
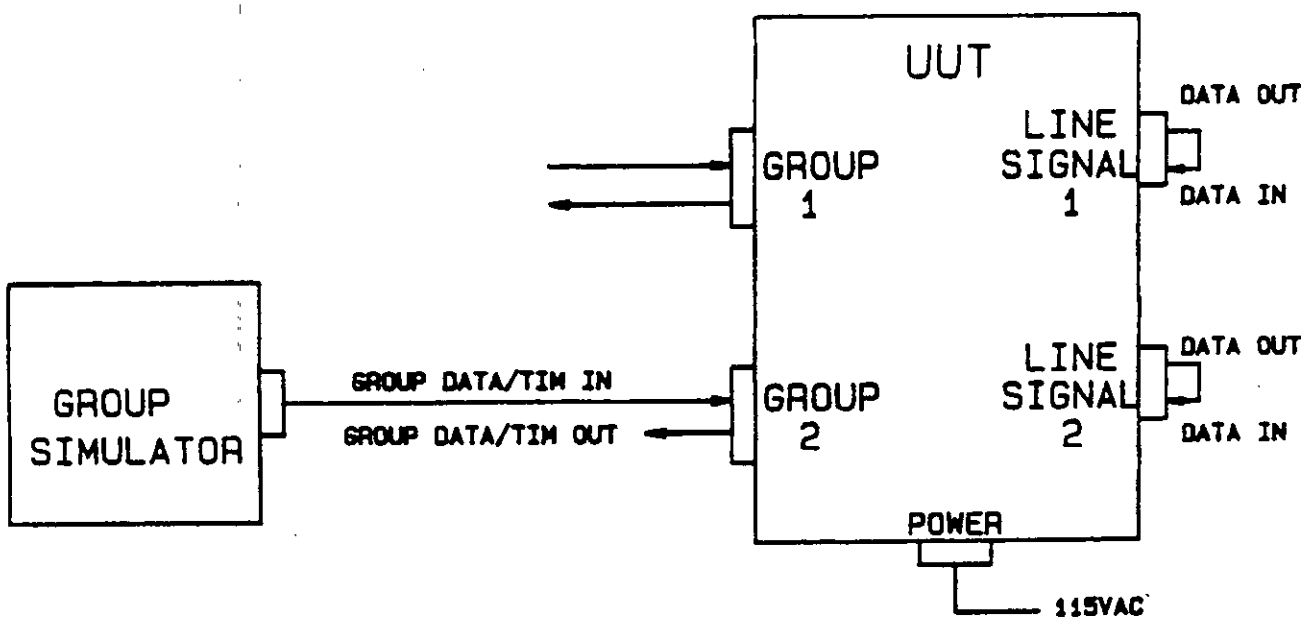



FIGURE 2. Line-to-line voltage levels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.



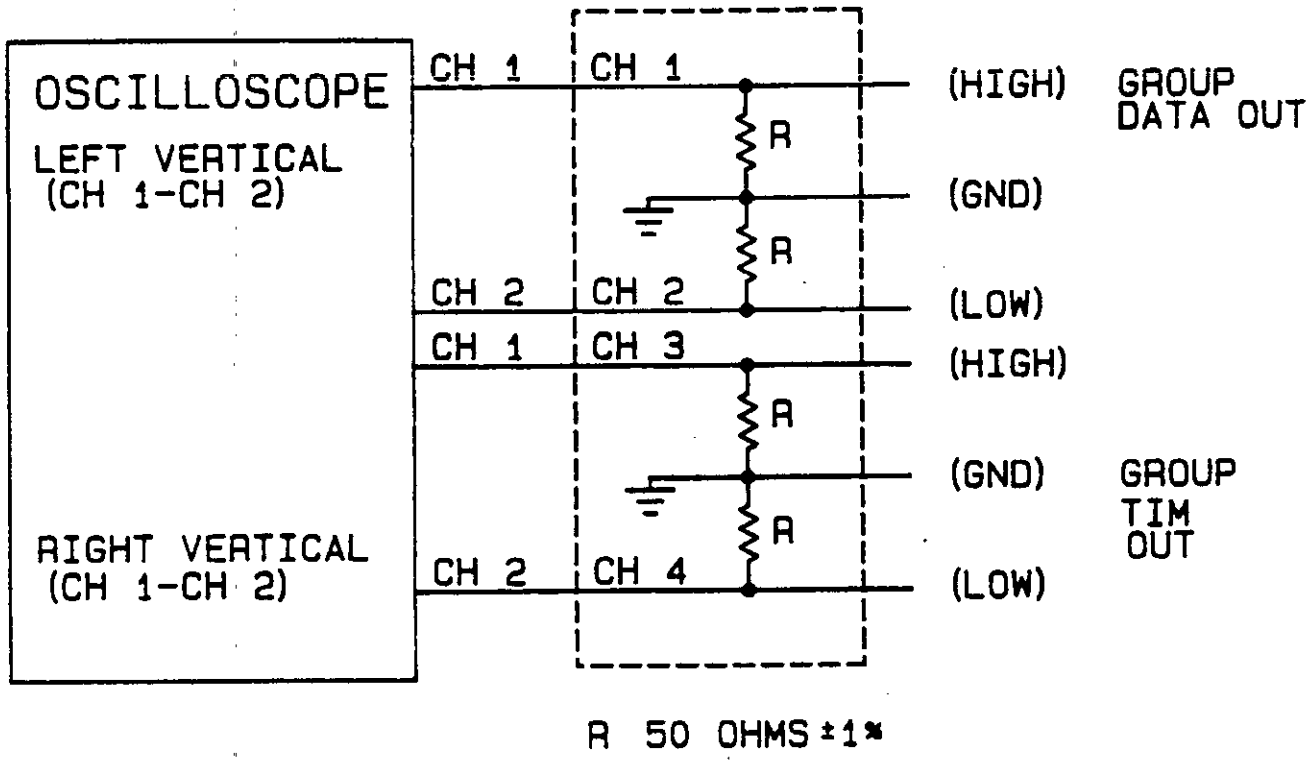




FIGURE 3. Clock/data phasing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

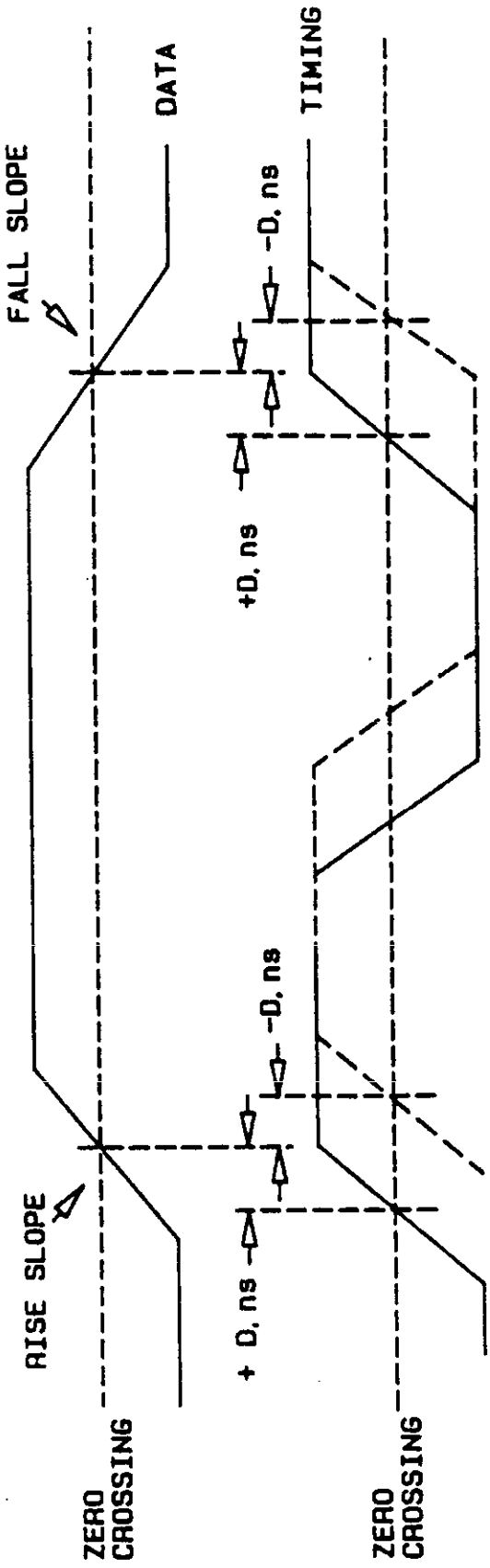



FIGURE 4 . Delay measurement (clock/data phasing).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

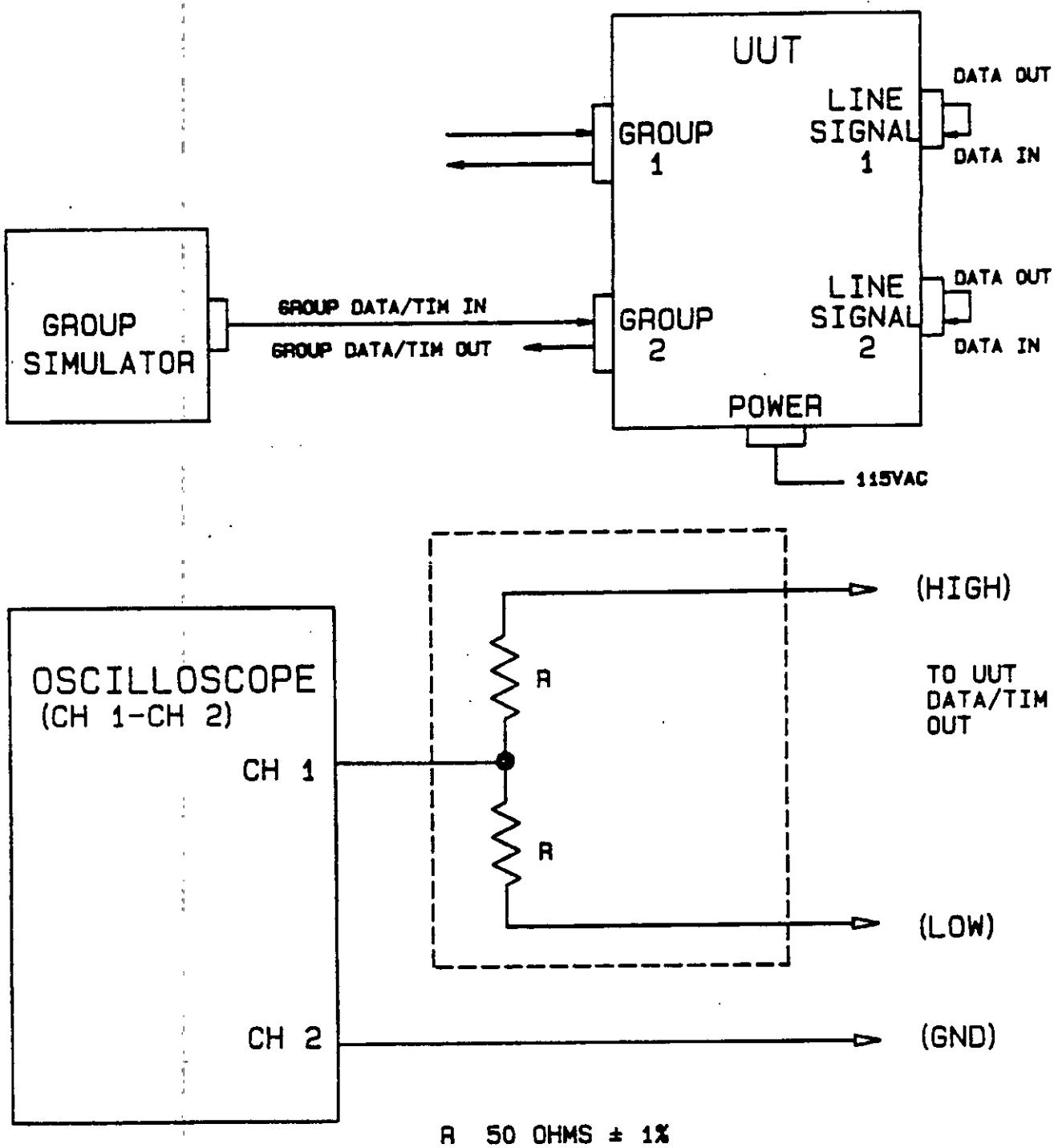
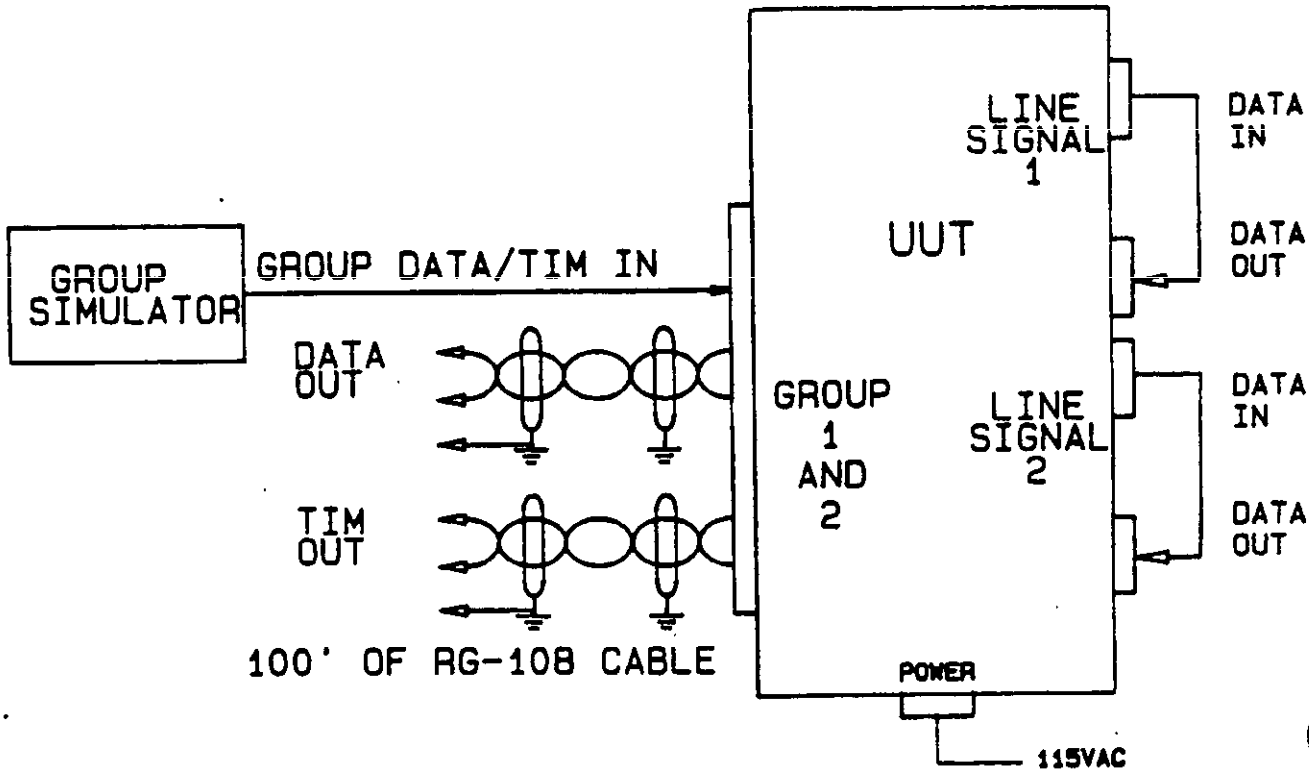
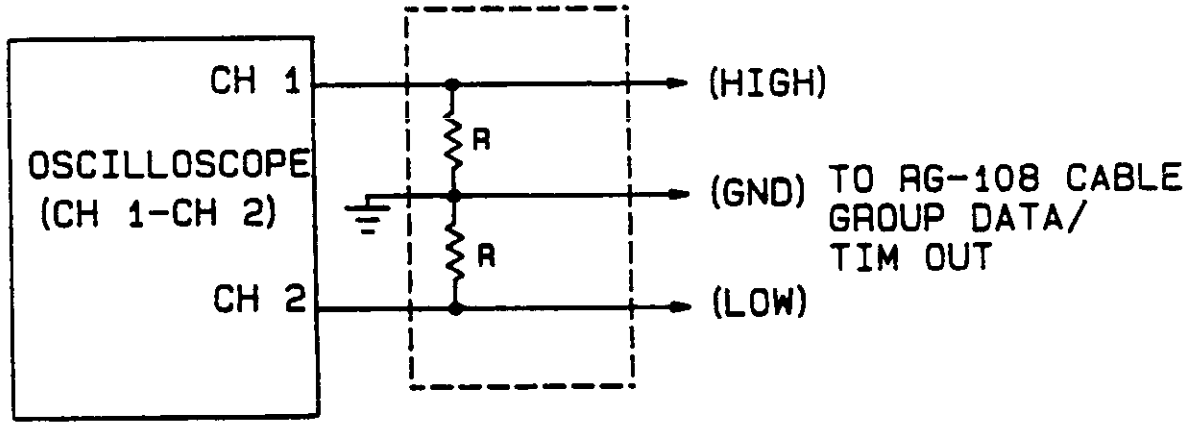



FIGURE 5. Offset.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.




R 39 OHMS ± 1%

FIGURE 6. Slew rate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

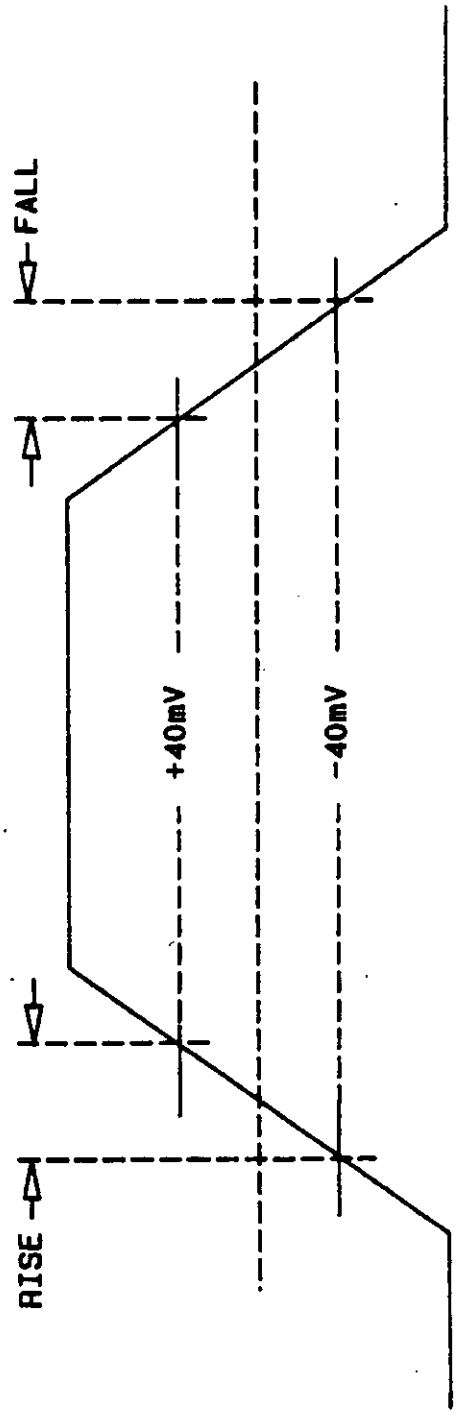



FIGURE 7. Rise and fall slopes (slew rate).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

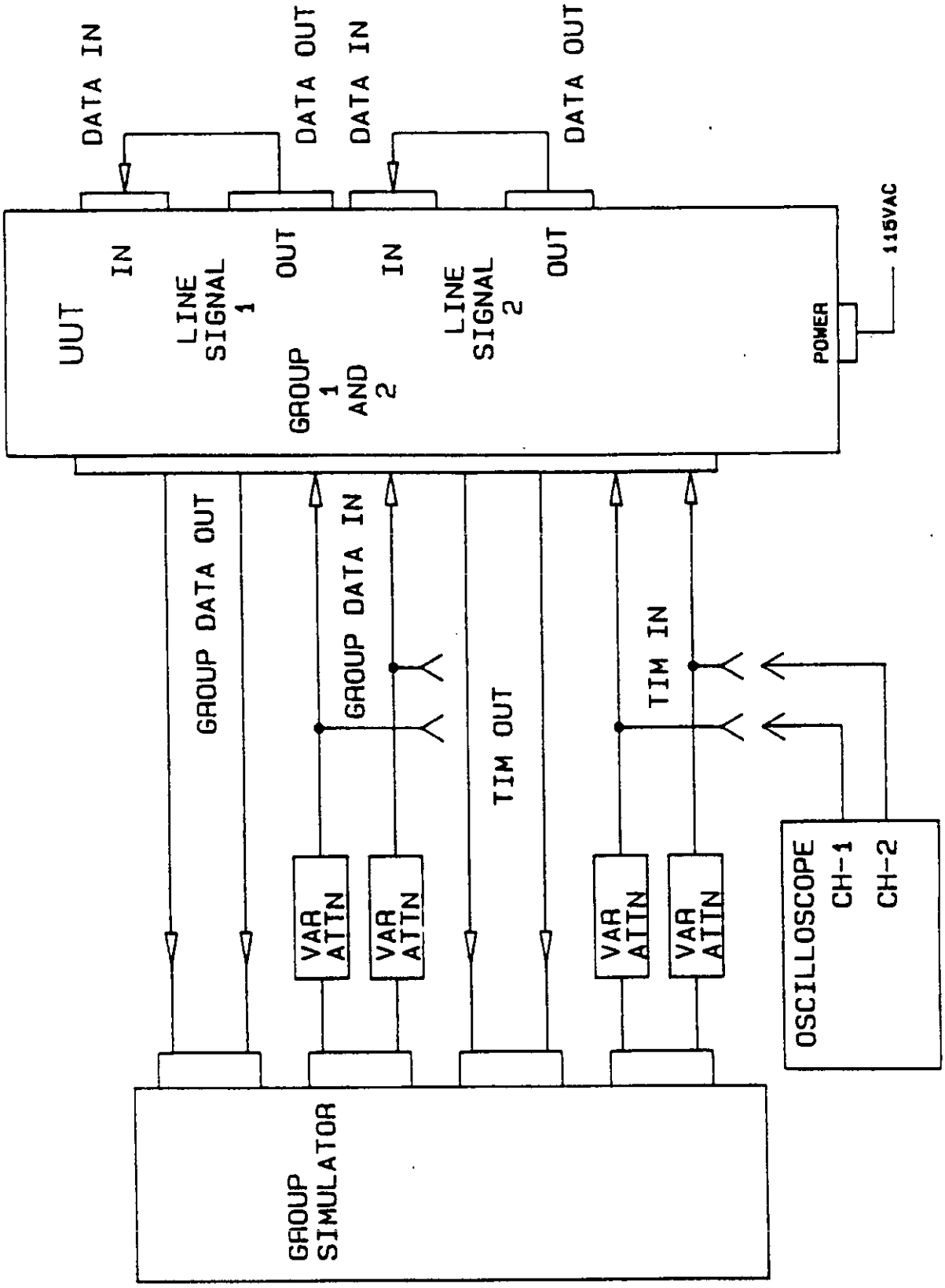



FIGURE 8. Receiver dynamic range.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

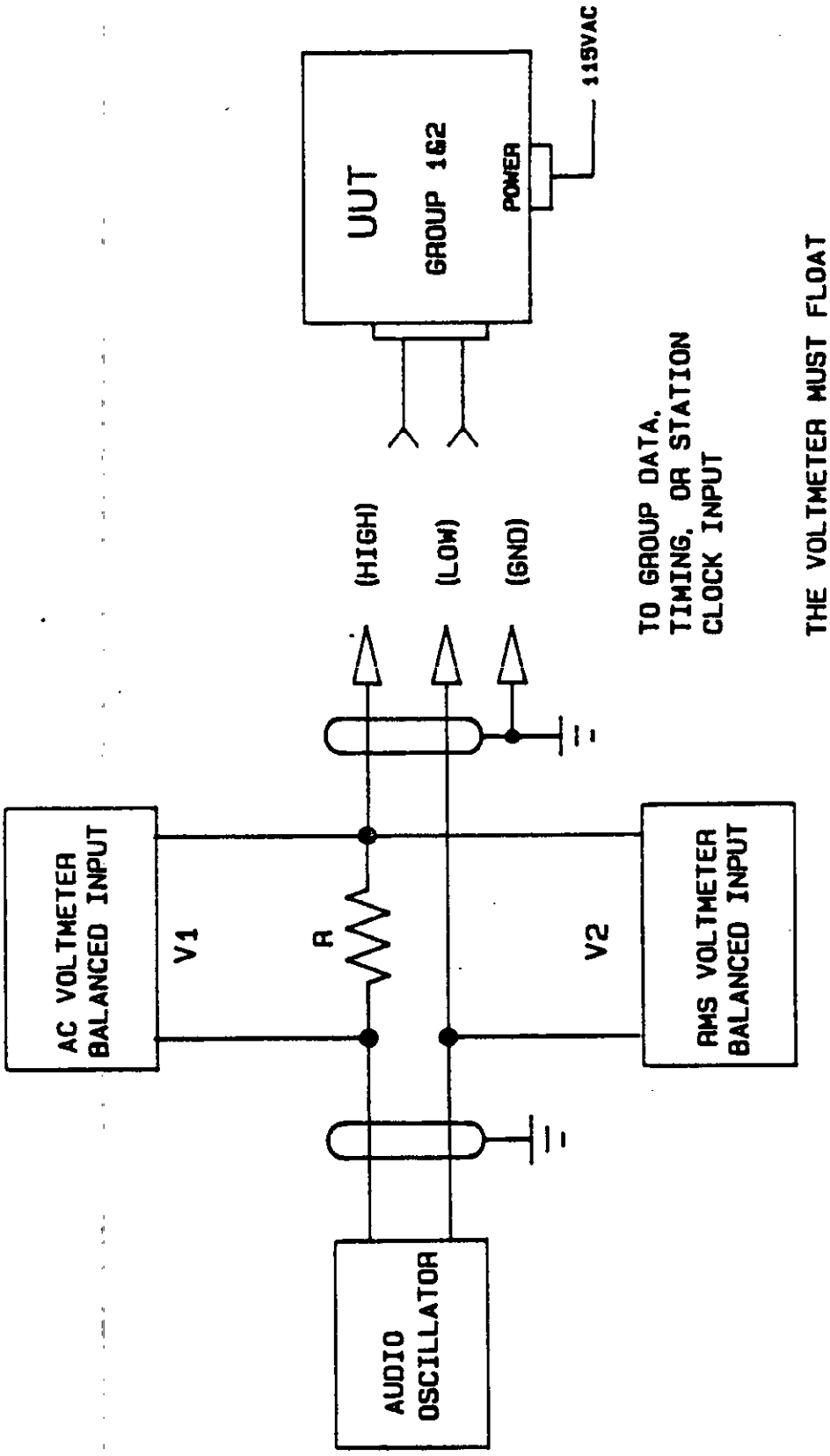



FIGURE 9. Input load impedance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

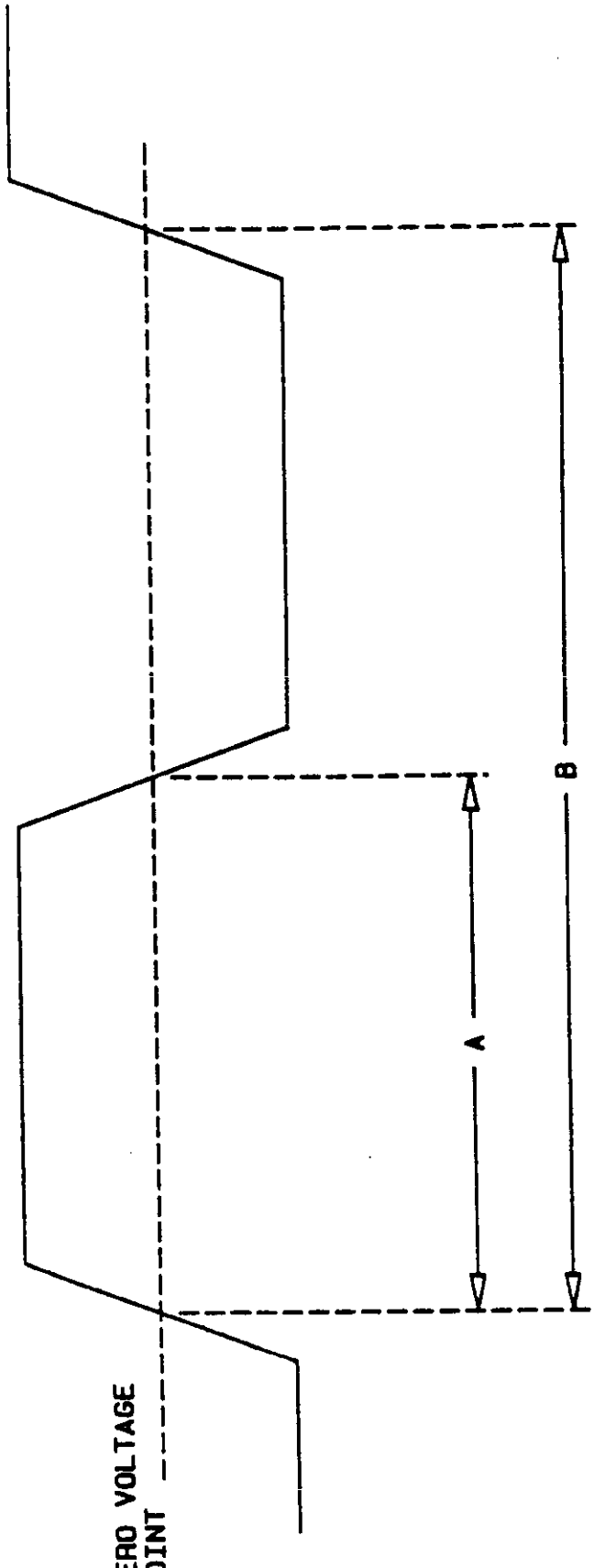
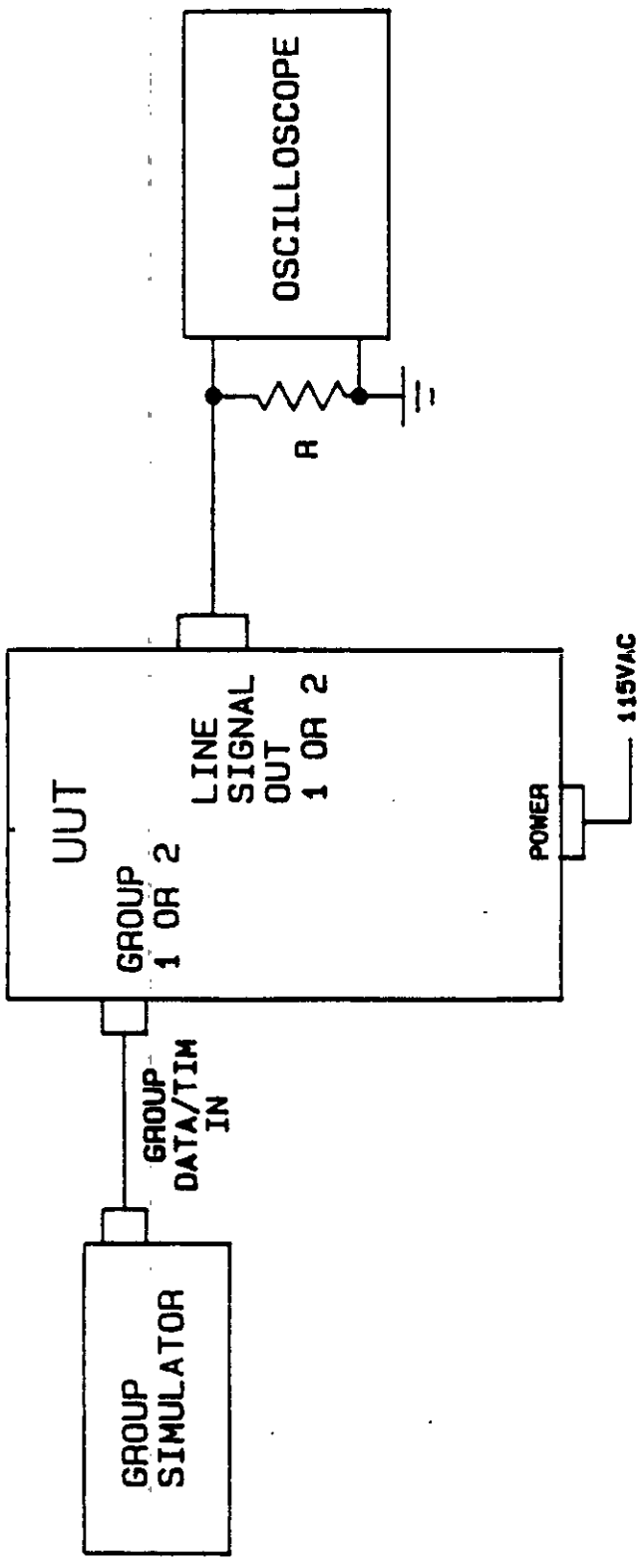



FIGURE 10. Clock symmetry.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.




R 57.6 OHMS ±1%

FIGURE 11. Output voltage levels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

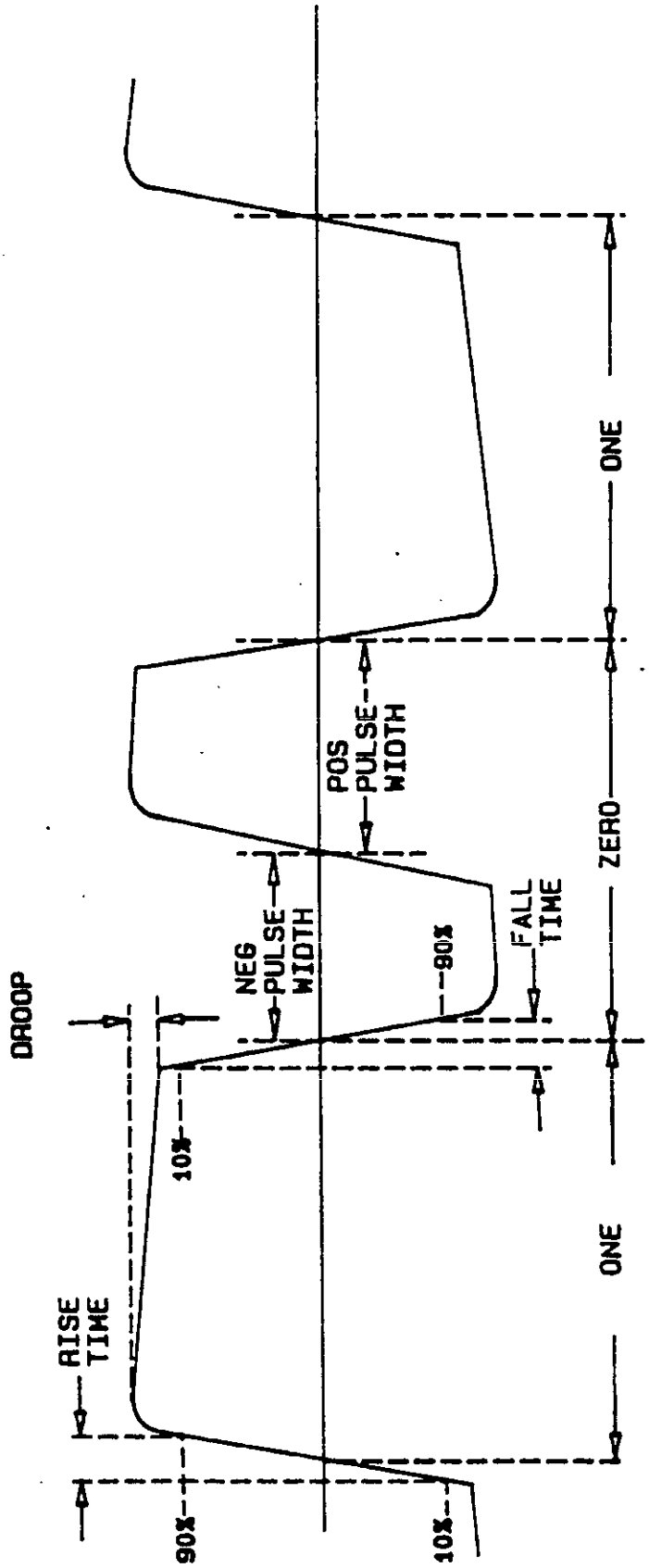
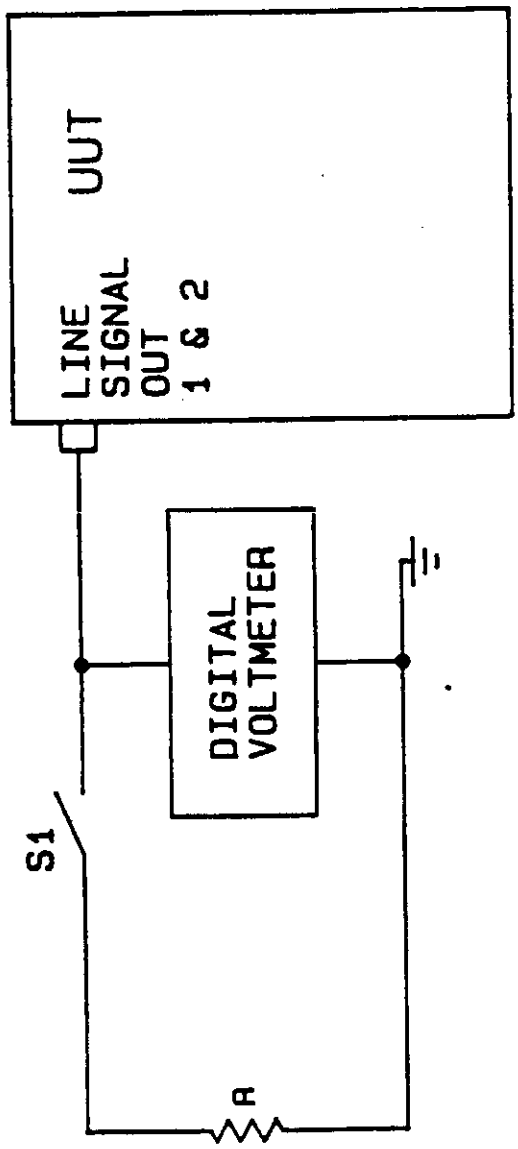



FIGURE 12. Diphase waveform (digital zero).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.




R 57.6 OHMS ± 1%

FIGURE 13. Cable side output impedance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

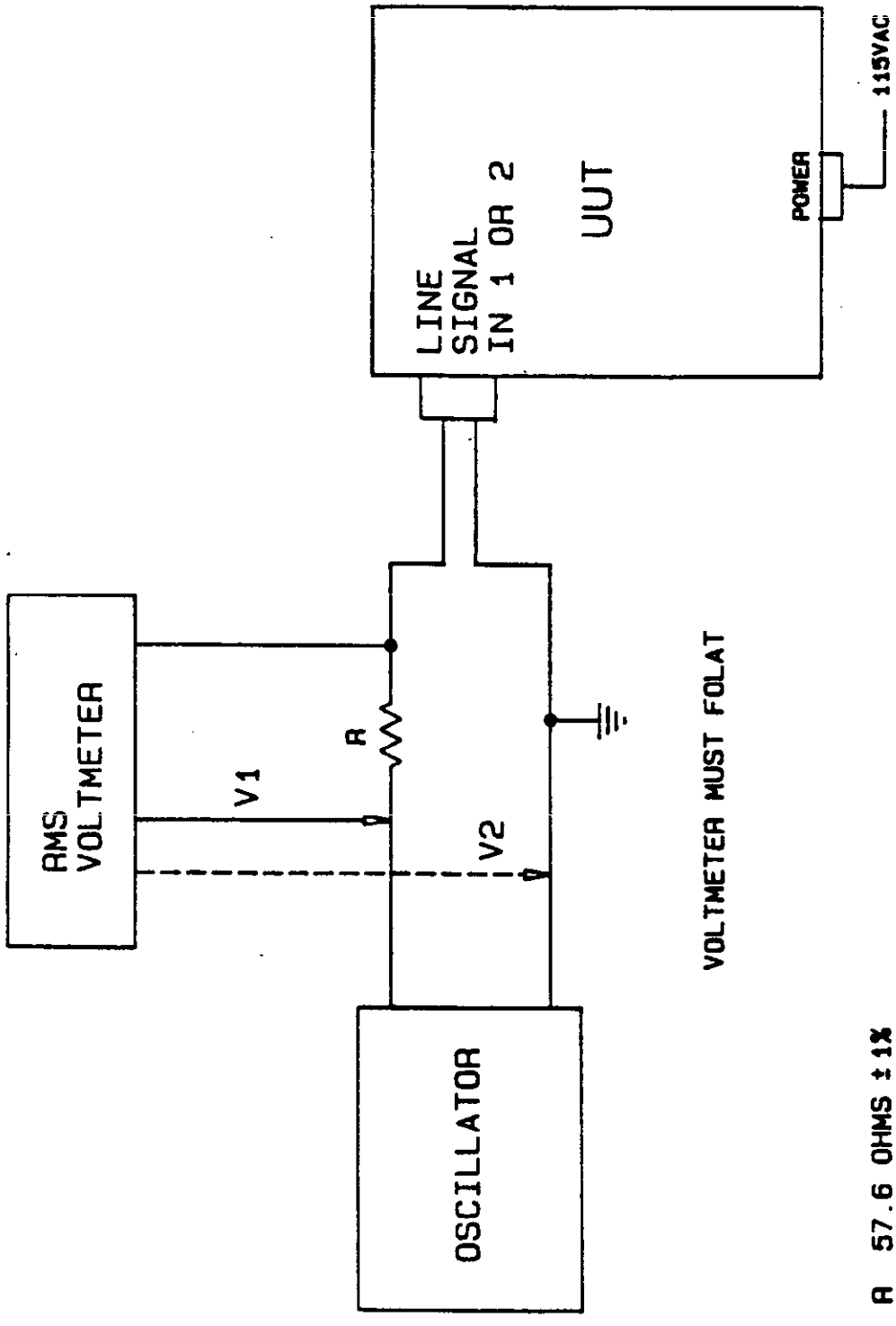



FIGURE 14. Cable side input impedance.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

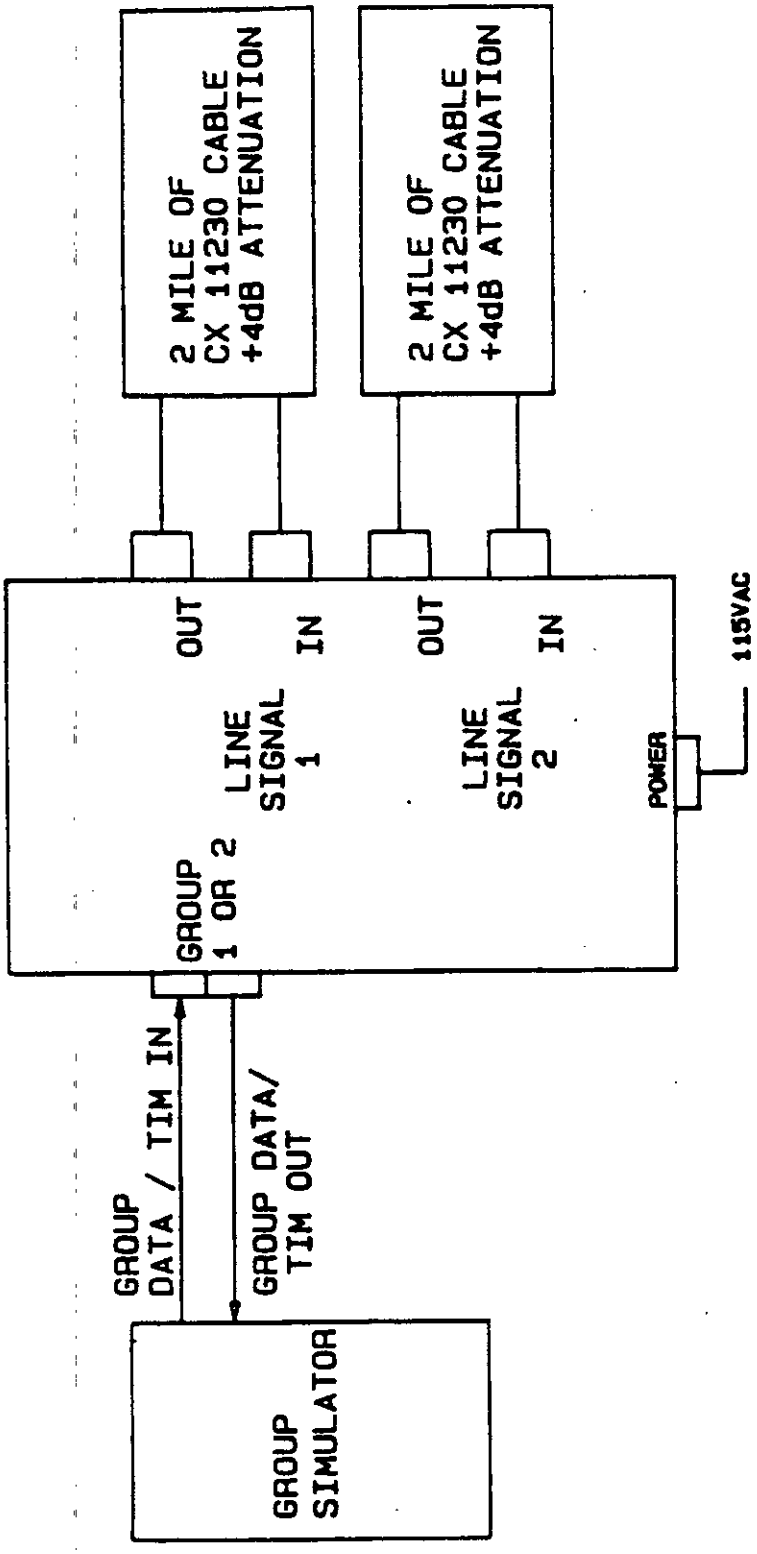



FIGURE 15. Receiver sensitivity.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

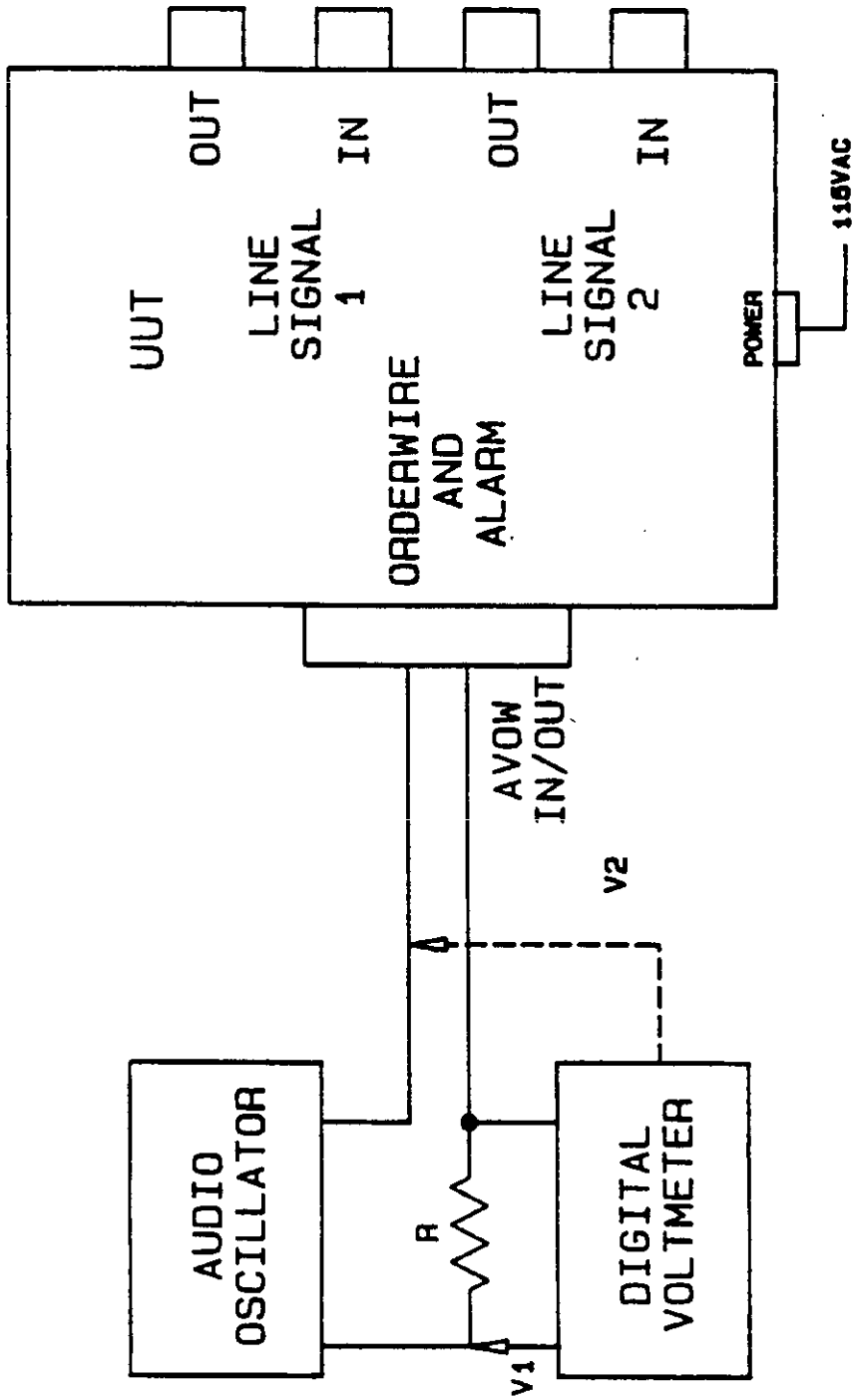
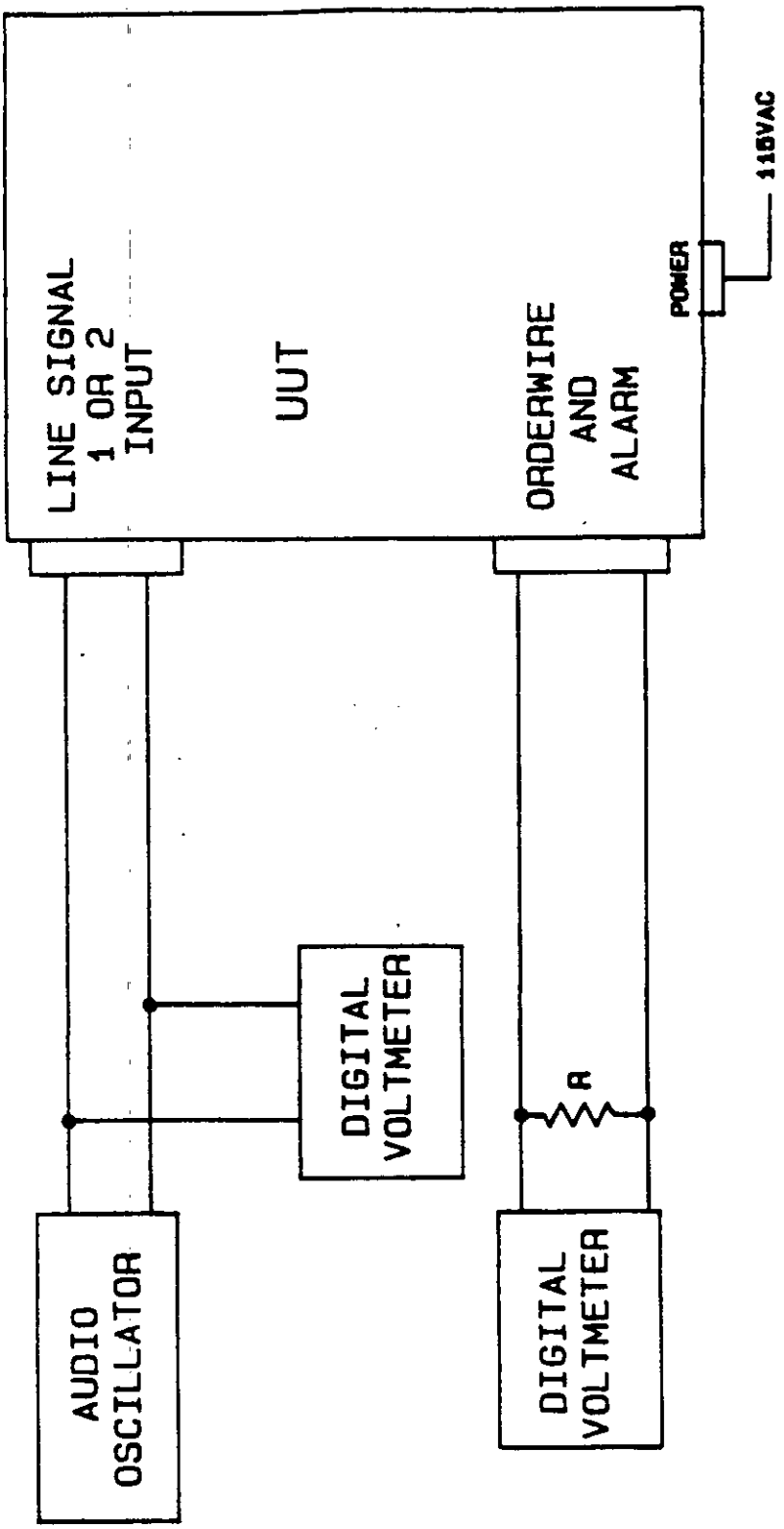



FIGURE 16. Orderwire input and output impedance.

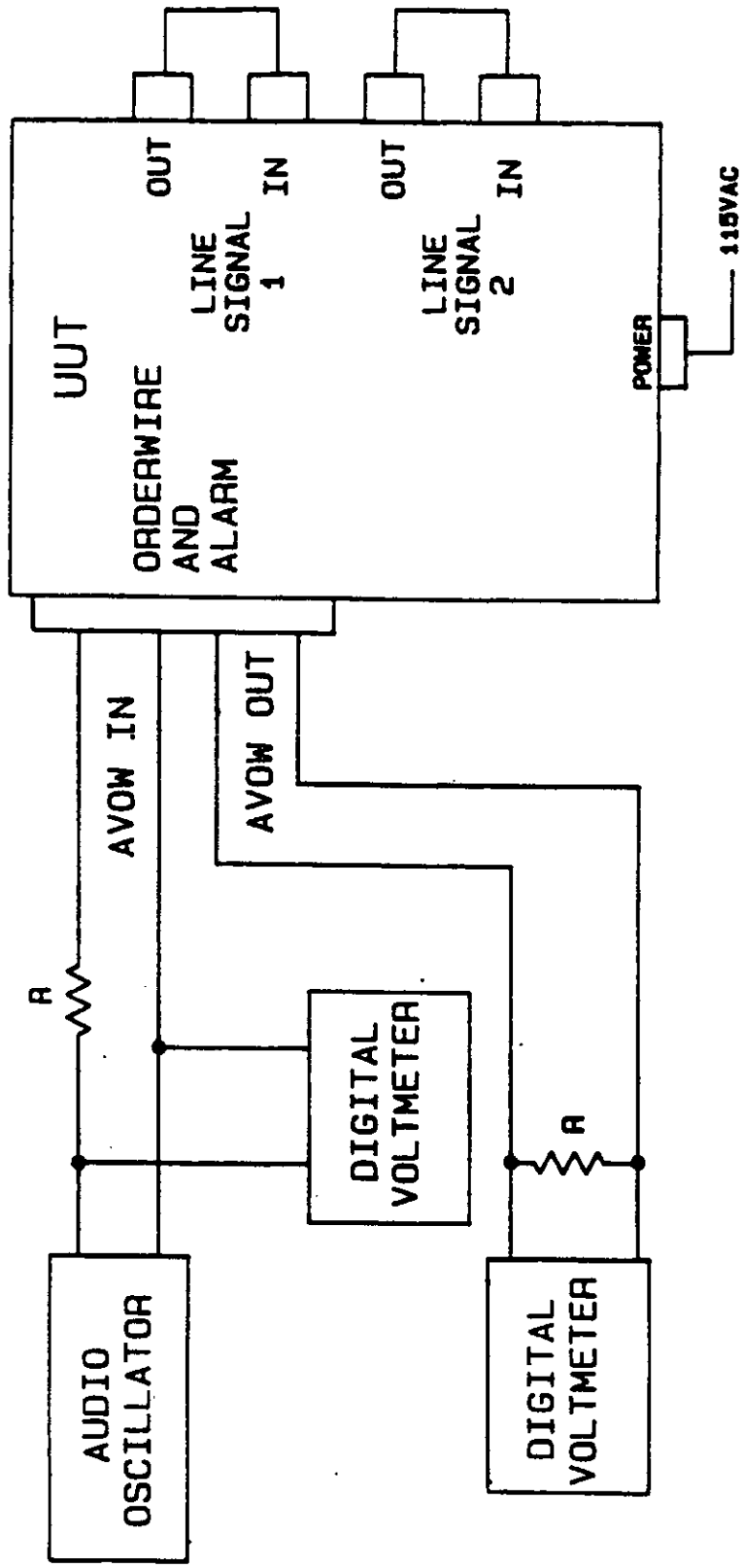
Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.




R 604 OHMS ± 1%

FIGURE 17. Equipment side maintenance orderwire level.

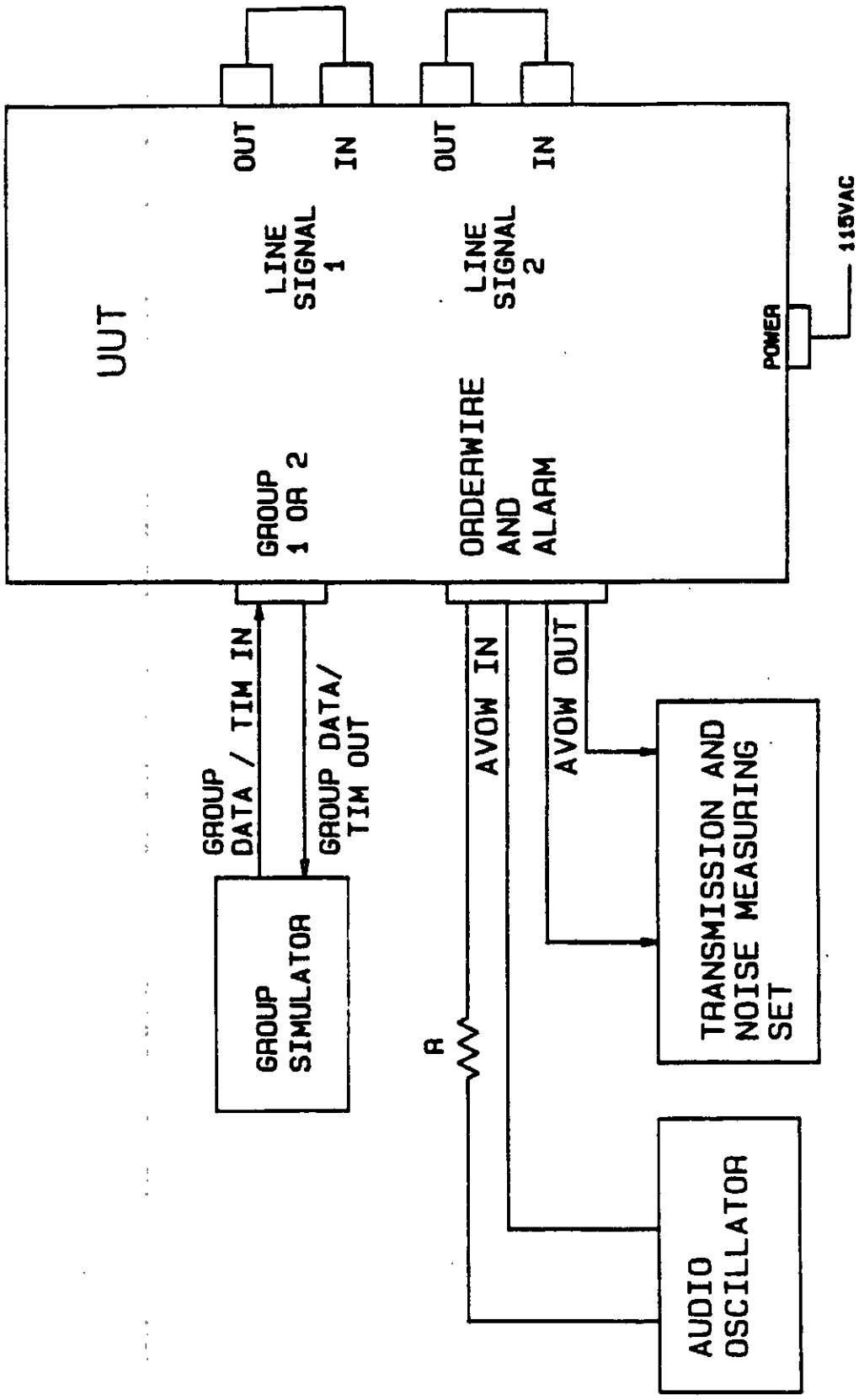
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.




R 604 OHMS ± 1%

FIGURE 18. Frequency response.

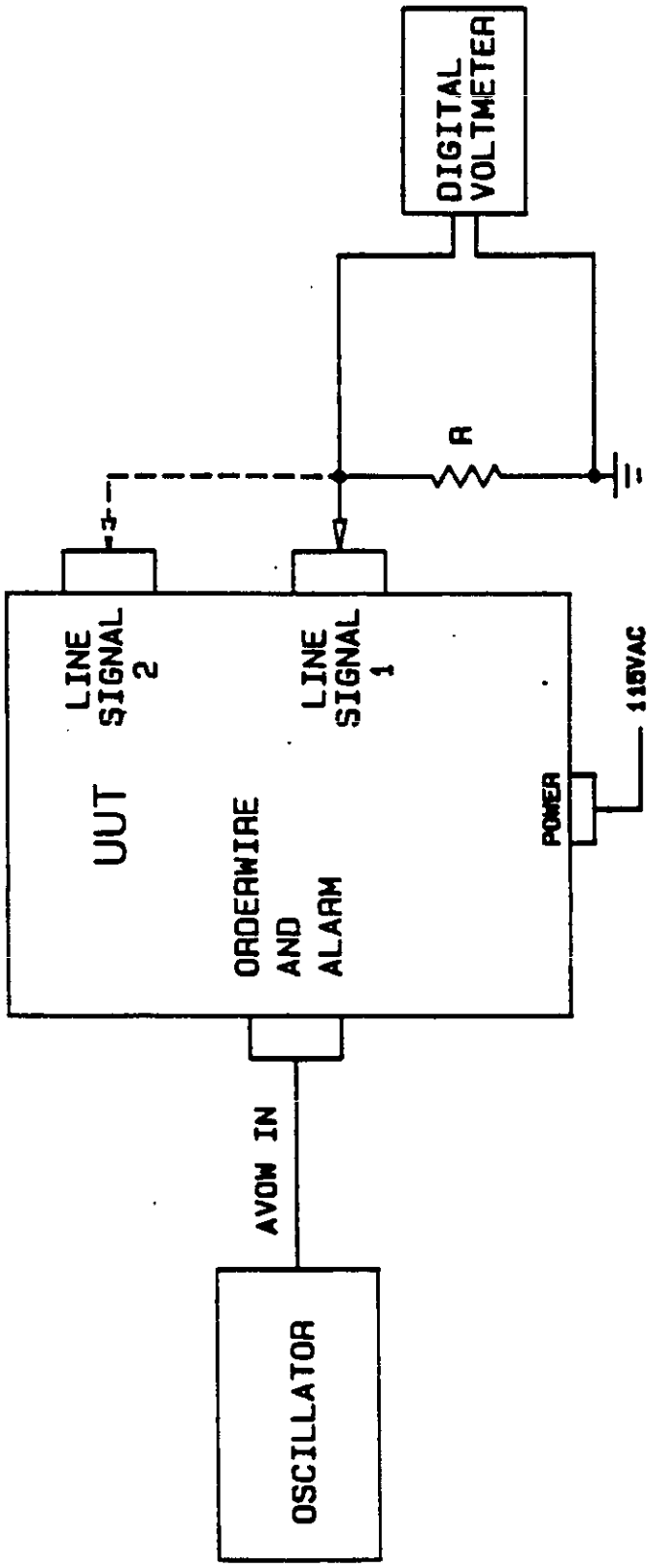
Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.




R 604 OHMS ± 1%

FIGURE 19. Maintenance orderwire signal-to-noise ratio.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.




R 57.6 OHMS ±1%

FIGURE 20. Orderwire output power level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

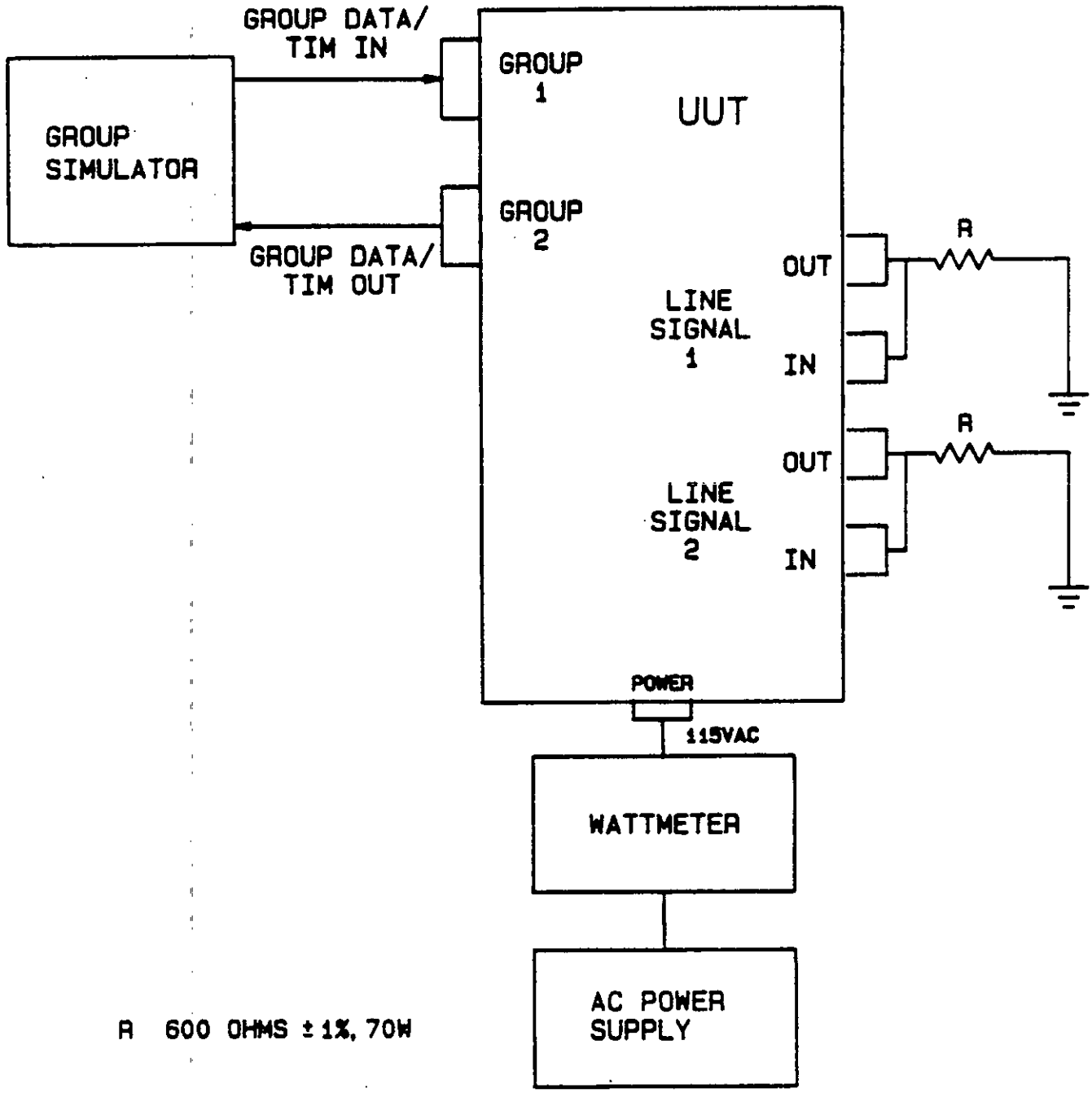
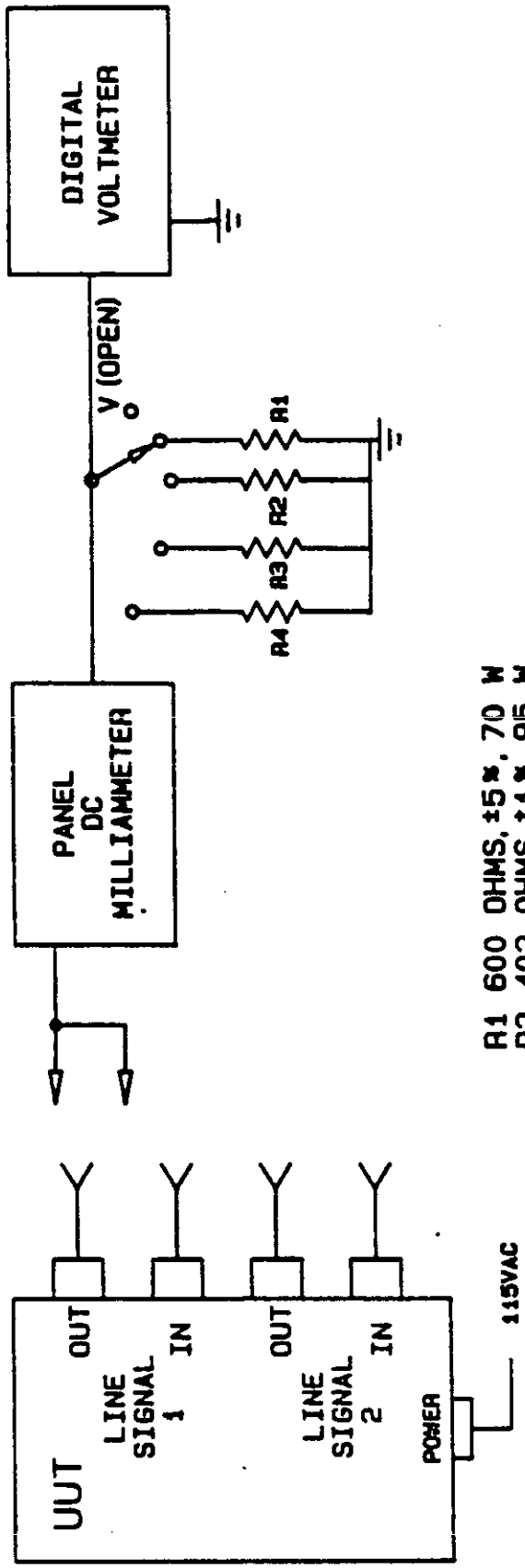



FIGURE 21. Prime power.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.




NOTE: Switch must be "MAKE BEFORE BREAK"

FIGURE 22. Power feed current limits.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

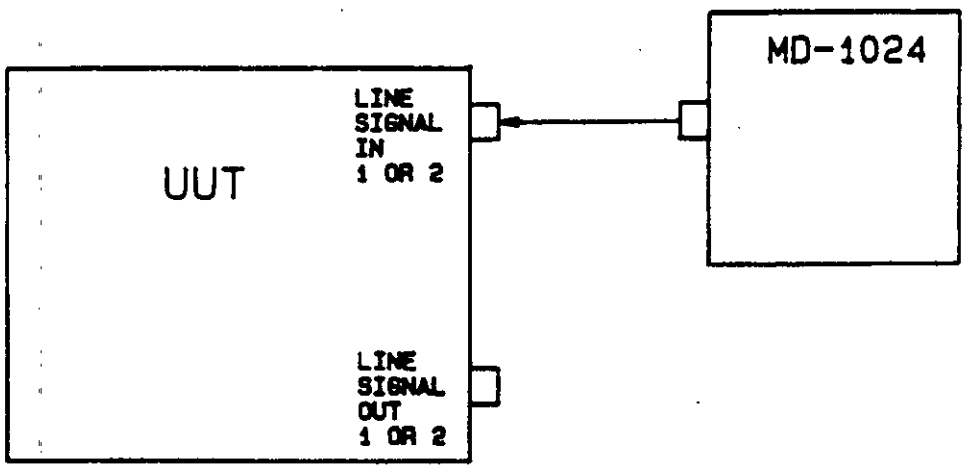



FIGURE 23. Current protection.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

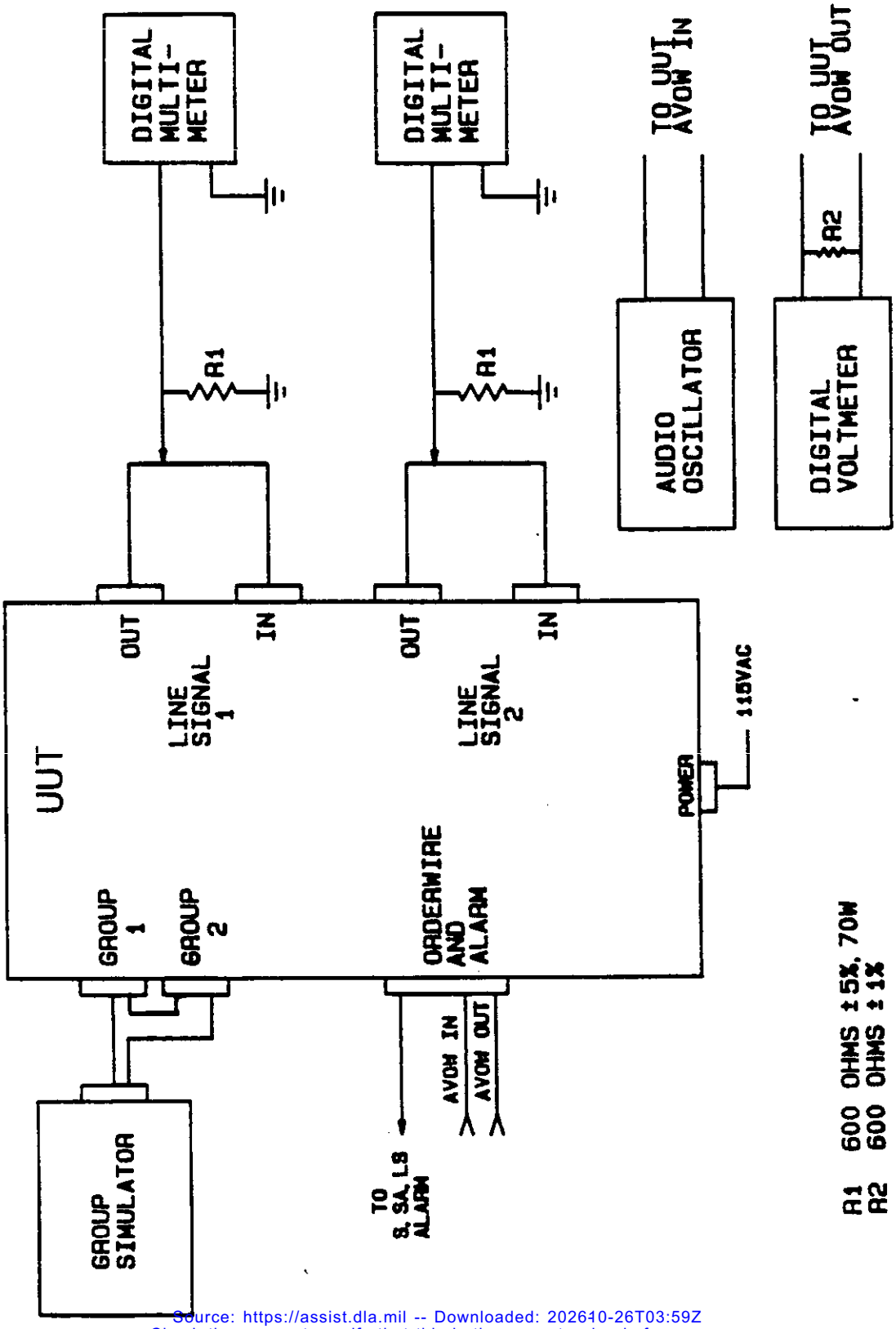



FIGURE 24. BITE and alarm operational tests.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

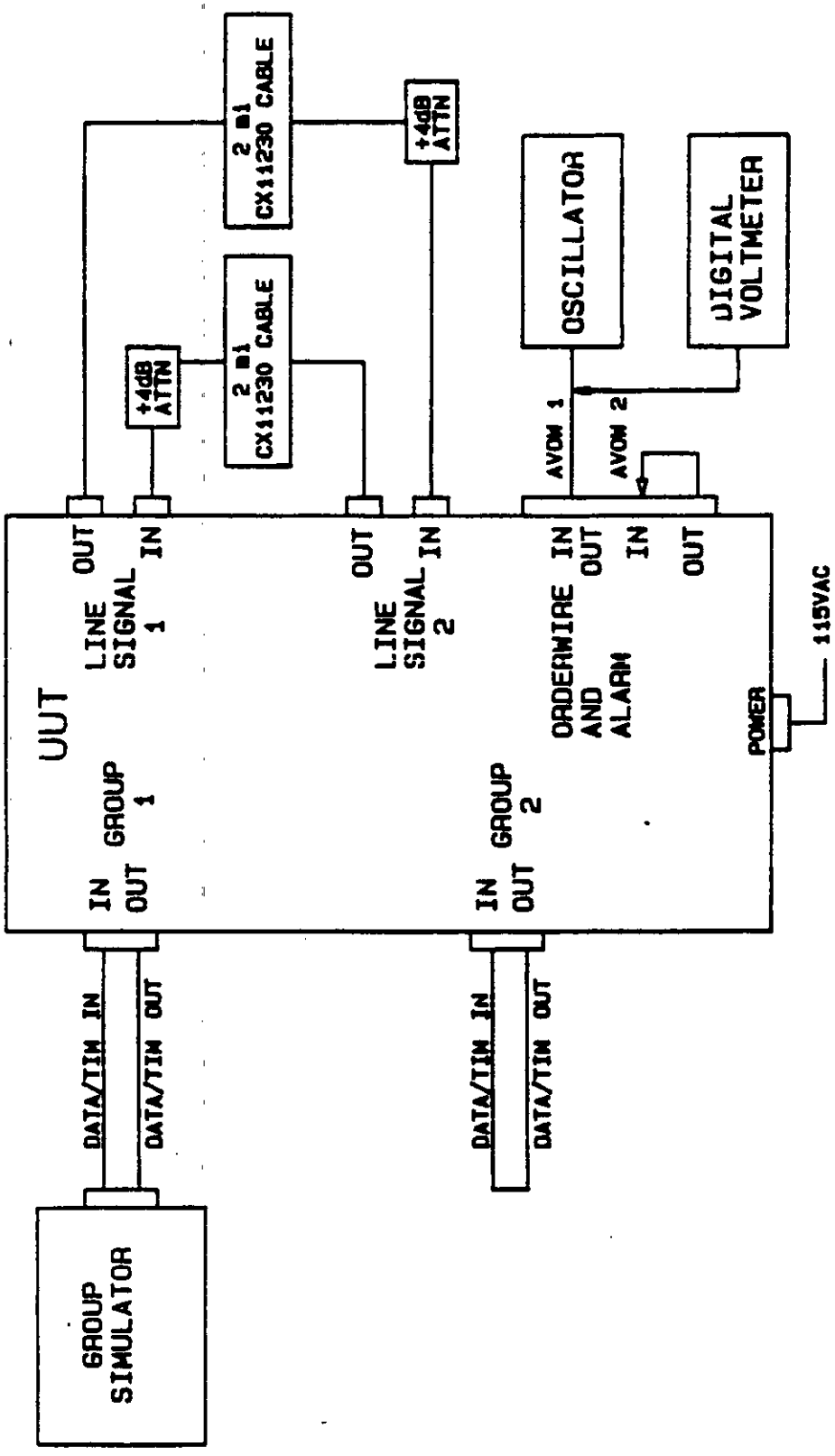



FIGURE 25. Full duplex signal detection and dynamic range.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

**INSTRUCTIONS:** In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

*(Fold along this line)*

*(Fold along this line)*

Cdr, CECOM
ATTN: AMSEL-ED-TO
Fort Monmouth, NJ 07703-9990

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 Alexandria, VA

POSTAGE WILL BE PAID BY

Commander
U.S. Army Communications-Electronics Command
and Fort Monmouth
ATTN: AMSEL-ED-TO
Fort Monmouth, NJ 07703-9990

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-M-49233A(CR) | Modem, Cable Drive, MD-1025/G |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.